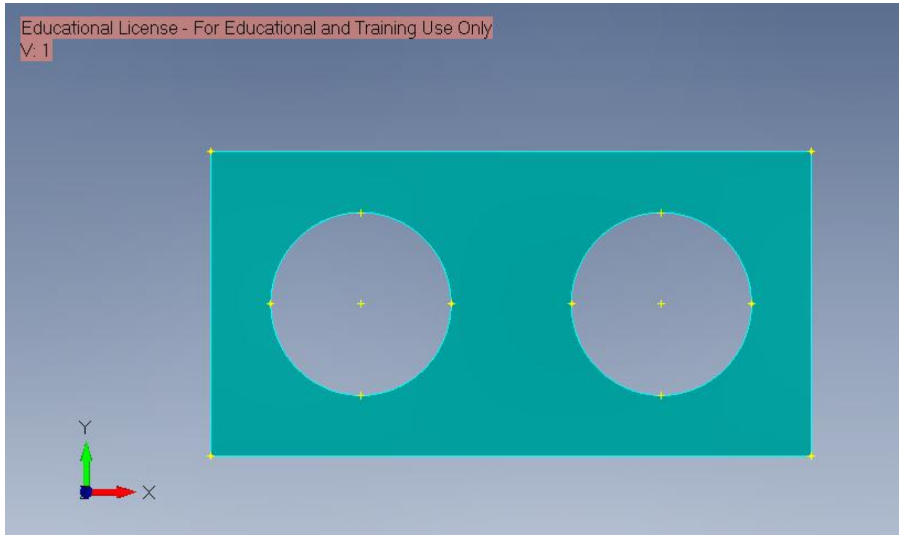
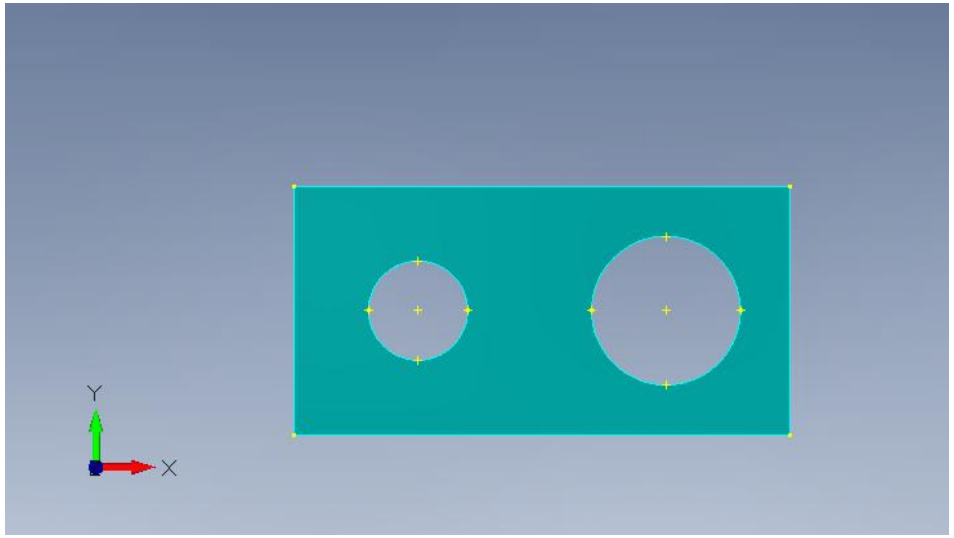
FEA Portfolio - FEMAP / Nastran

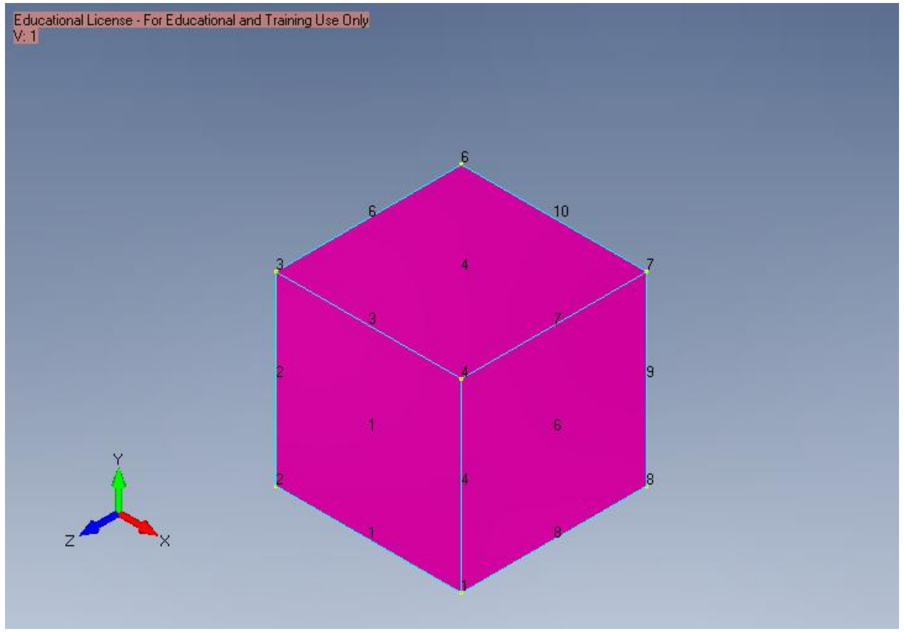
Gustabo Lazaro

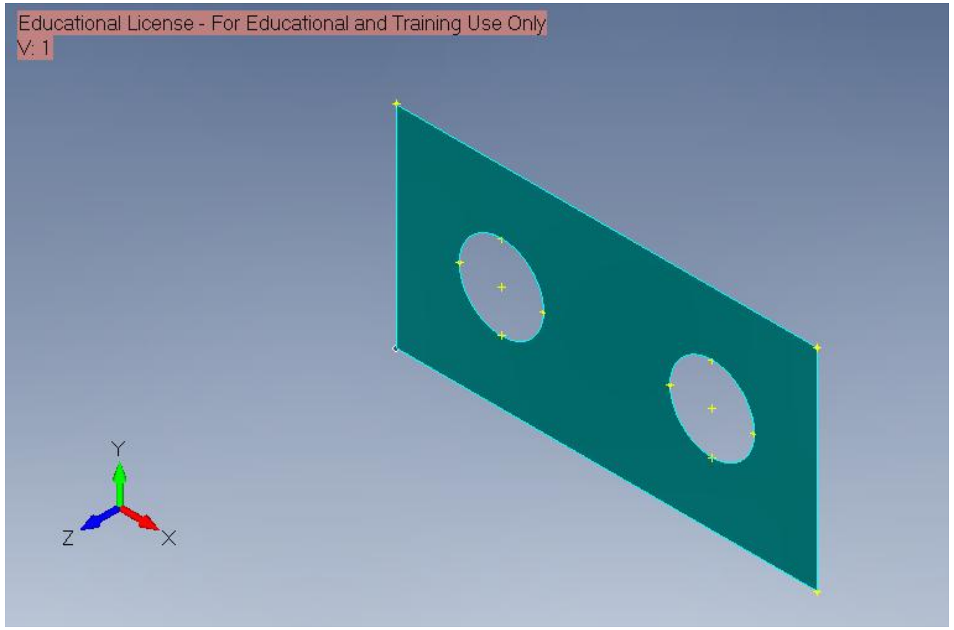
Lab 1 3D CAD – Geometry





Created basic 2D and 3D geometry in FEMAP, including an open-hole plate and a unit cube.

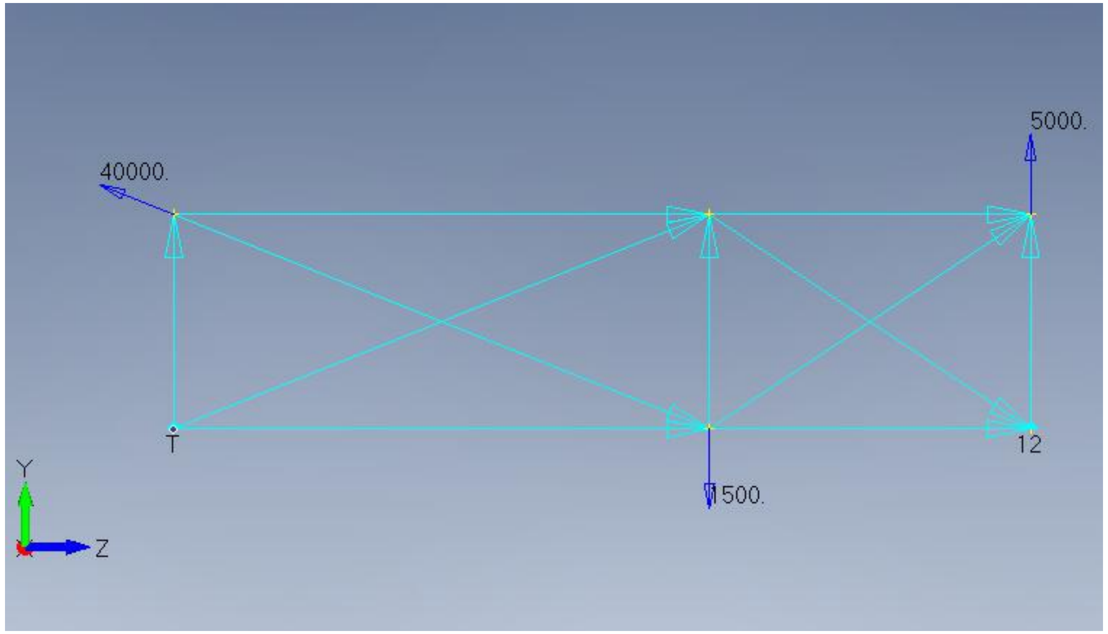




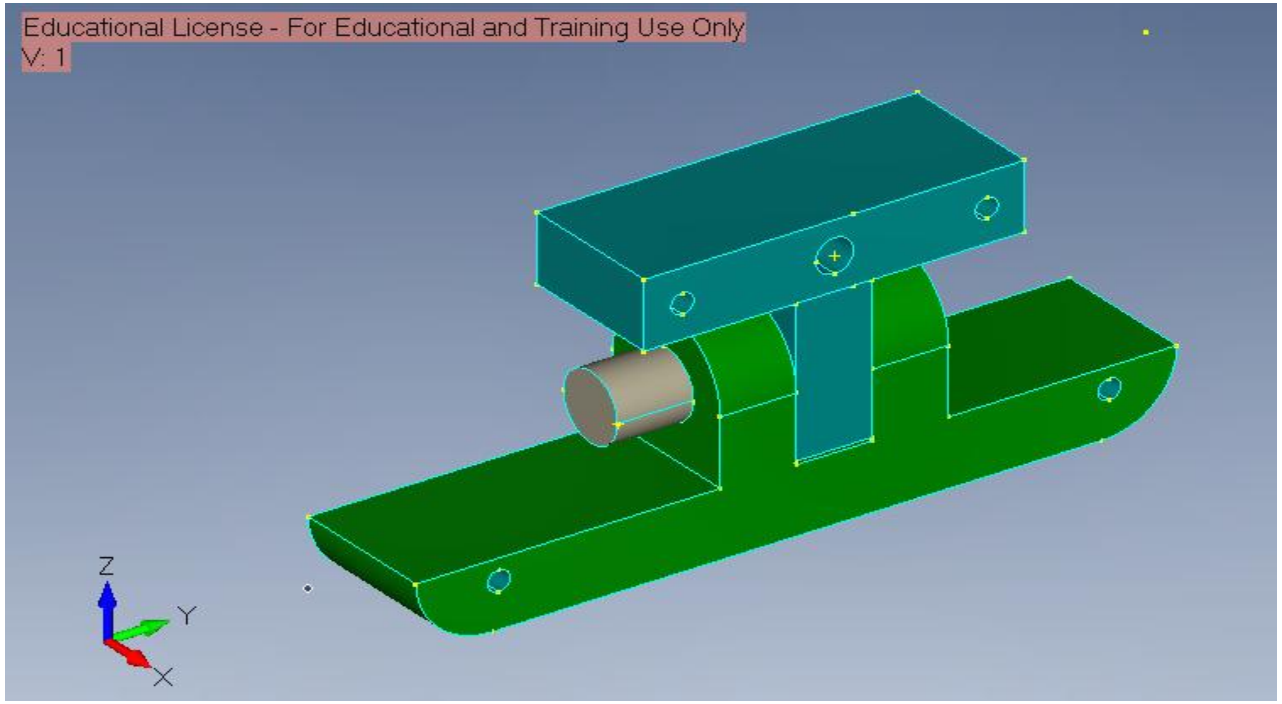
Practiced using curves, surfaces, trimming tools, workplanes, and grid snapping to build clean geometry.

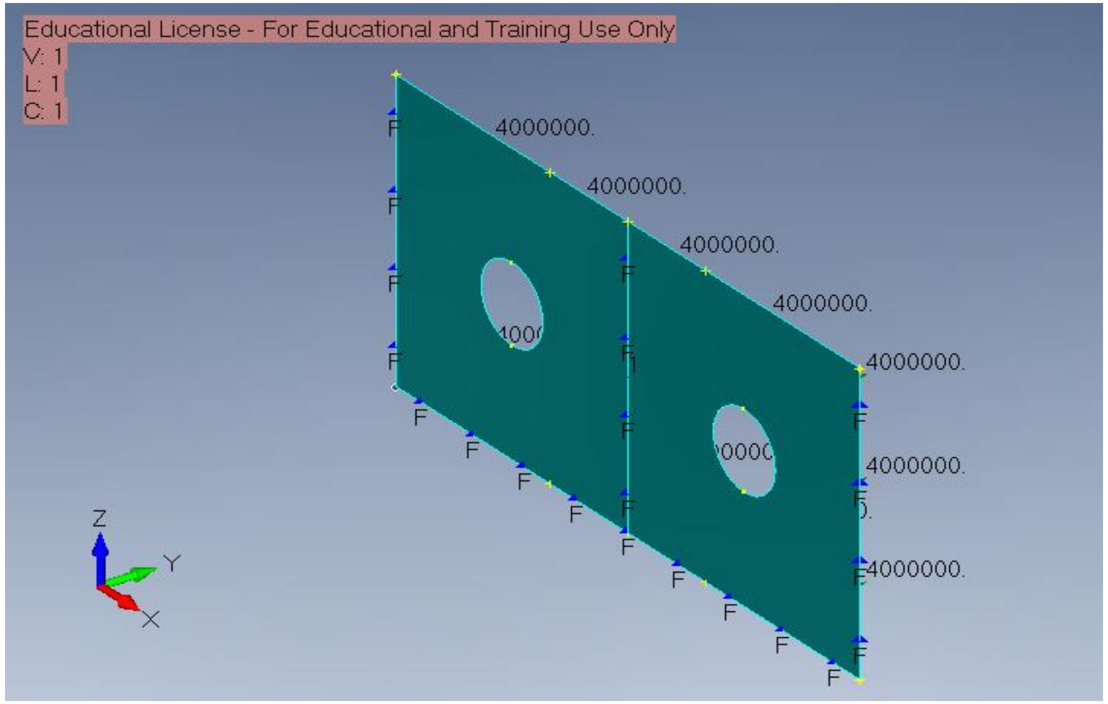
Lab 2 1D, 2D, 3D CAD — More on Geometry and LBC's

Created the planar truss geometry on the YZ plane using points and line segments.
Applied the required loads and boundary conditions.



Modeled a 3D base, pin, and arm assembly in FEMAP using solid geometry tools.

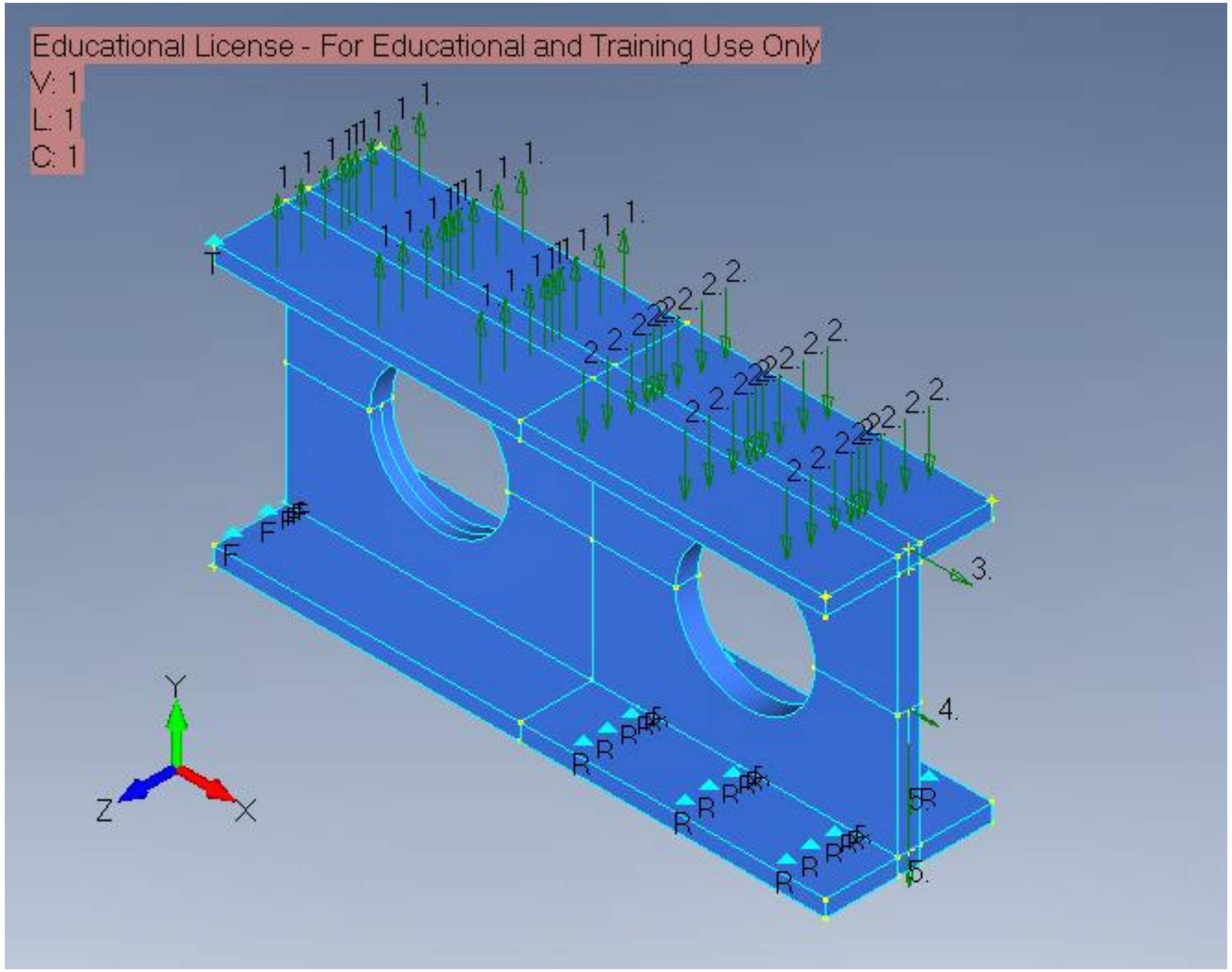




Lab 3 3D CAD Solid Models

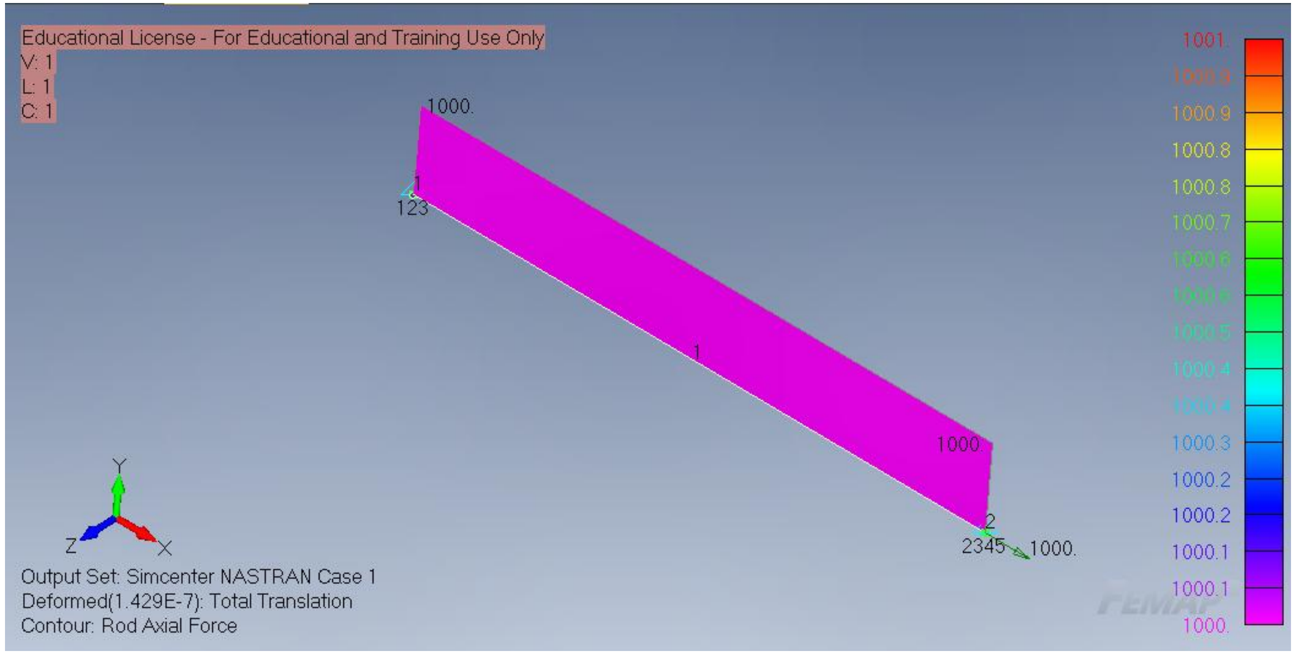
Modeled the rib structure as a 3D solid part with top and bottom flanges, web thickness, and two circular holes.

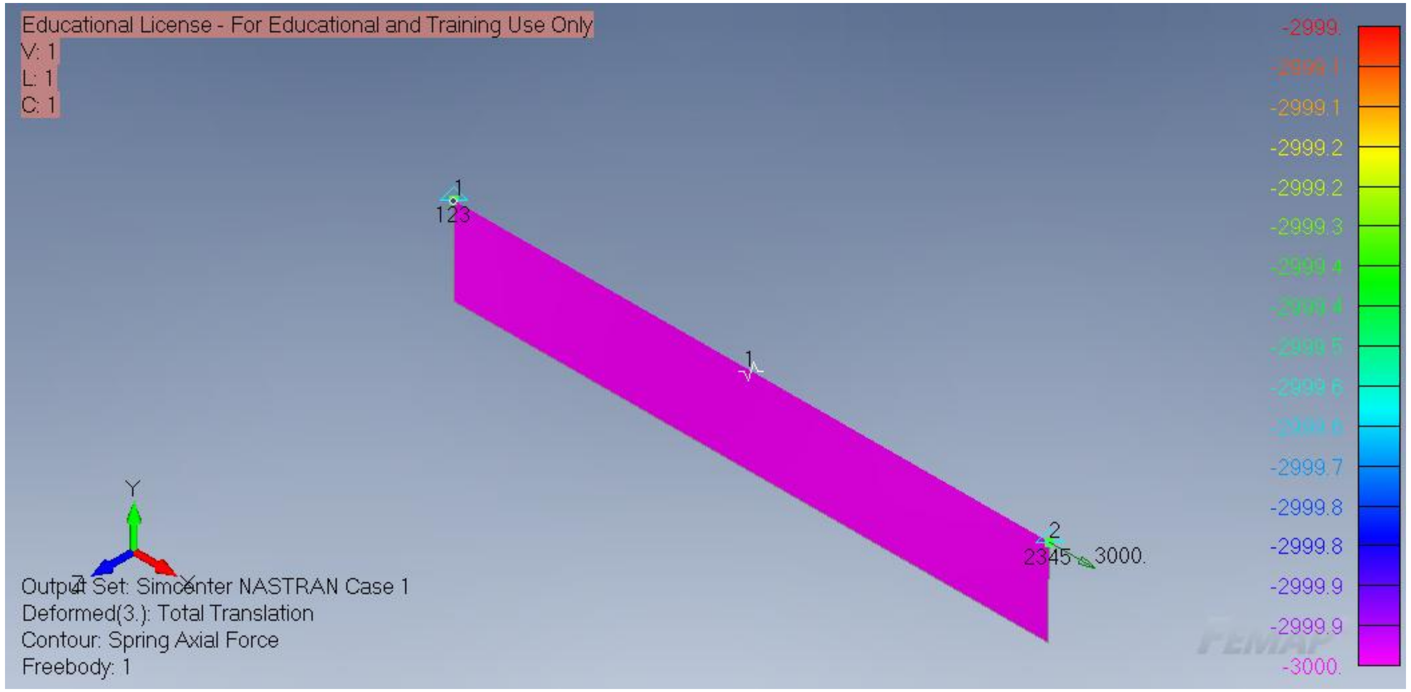
Defined surface loads and support constraints based on the provided loading and boundary conditions.



Lab 4 1D Spring and 1D Rod Elements

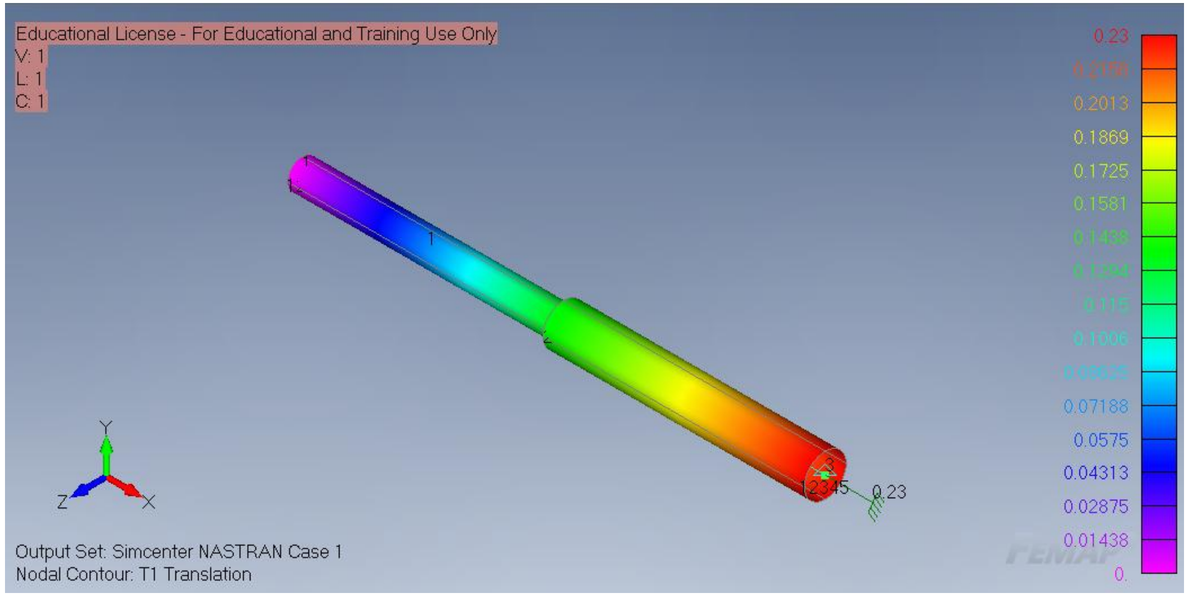
Created a 1D spring model in FEMAP and assigned the required spring stiffness.
I ran the analysis to determine the spring displacement and internal element force.



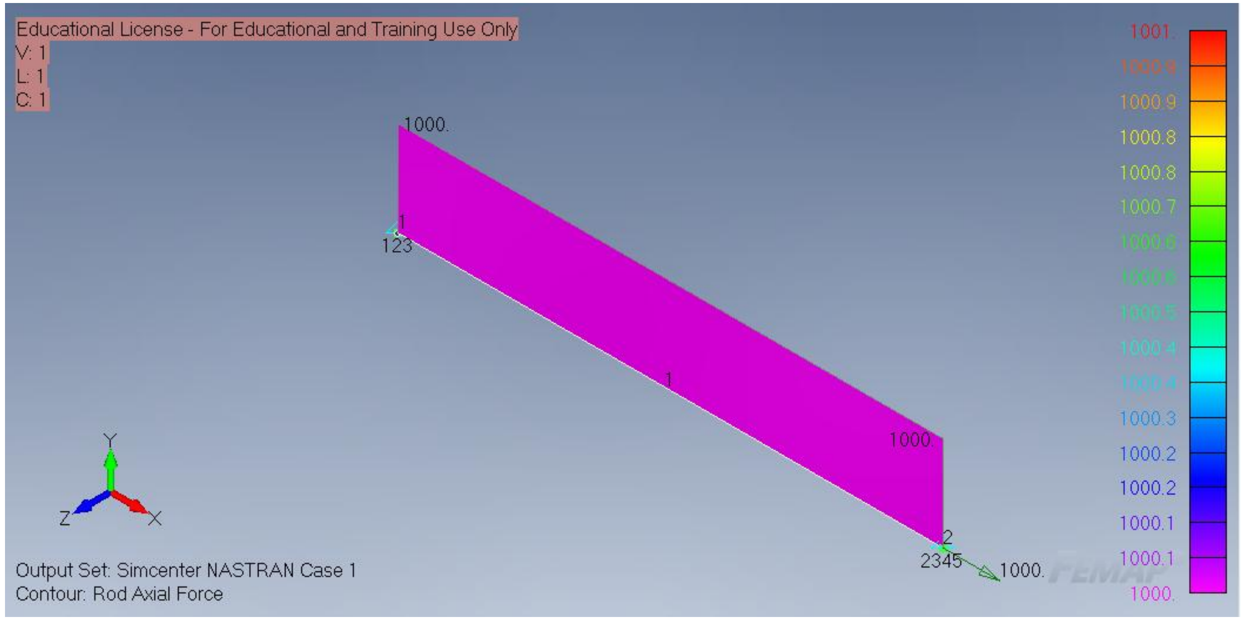


Modeled a 1D aluminum rod with assigned material properties, cross-sectional area, supports, and axial loading.

Lab 5 1D Spring/Mass and 1D Rod from Geometry

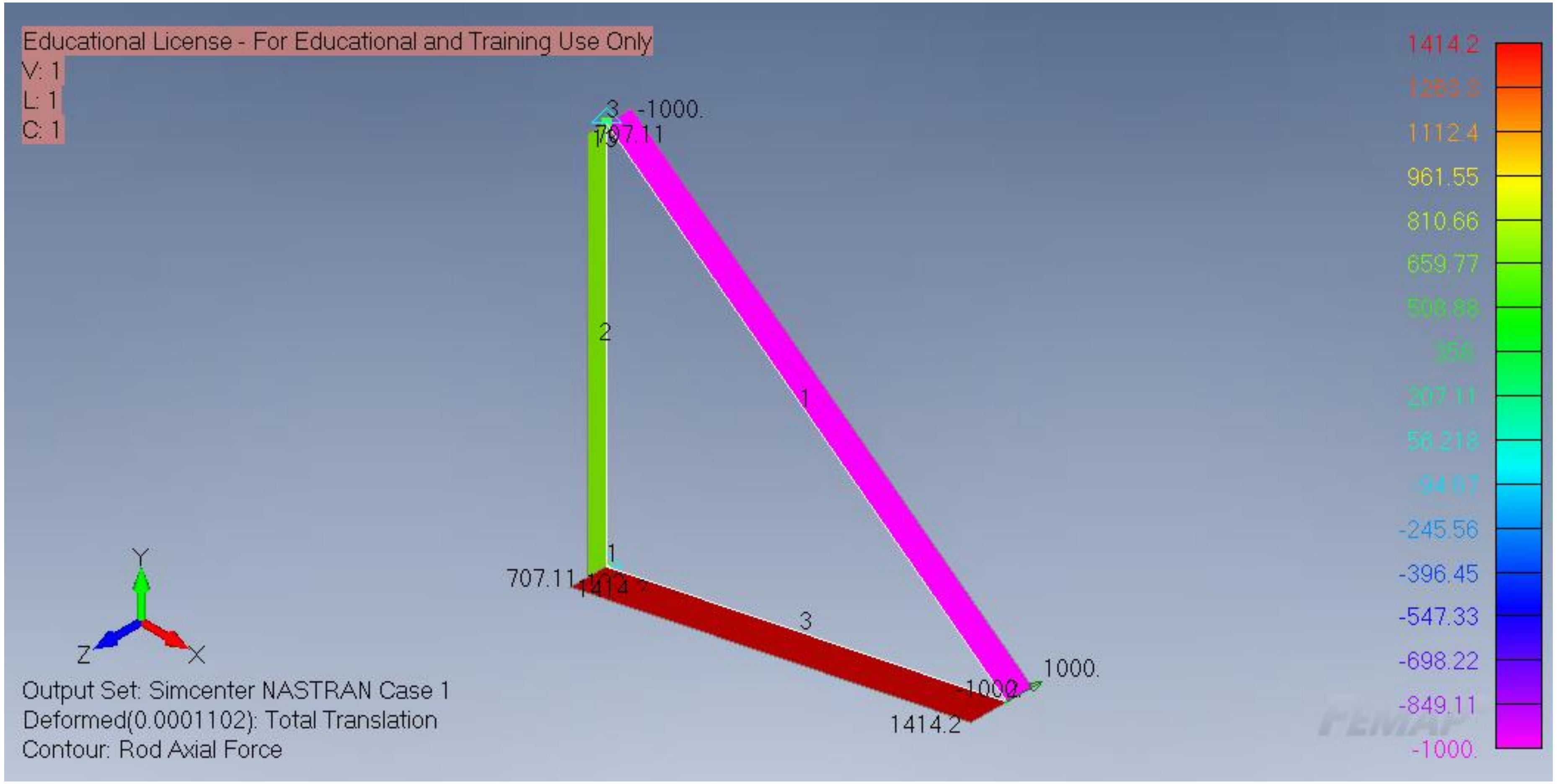


Modeled a 1D rod and applied displacement type loading at the end node.
I ran the analysis to determine nodal displacements, element forces, stresses, and support reactions.



Created a 1D spring model with an aluminum spring stiffness and a hanging mass load.
Analyzed the spring response to determine the nodal displacement and internal spring force caused by the hanging mass.

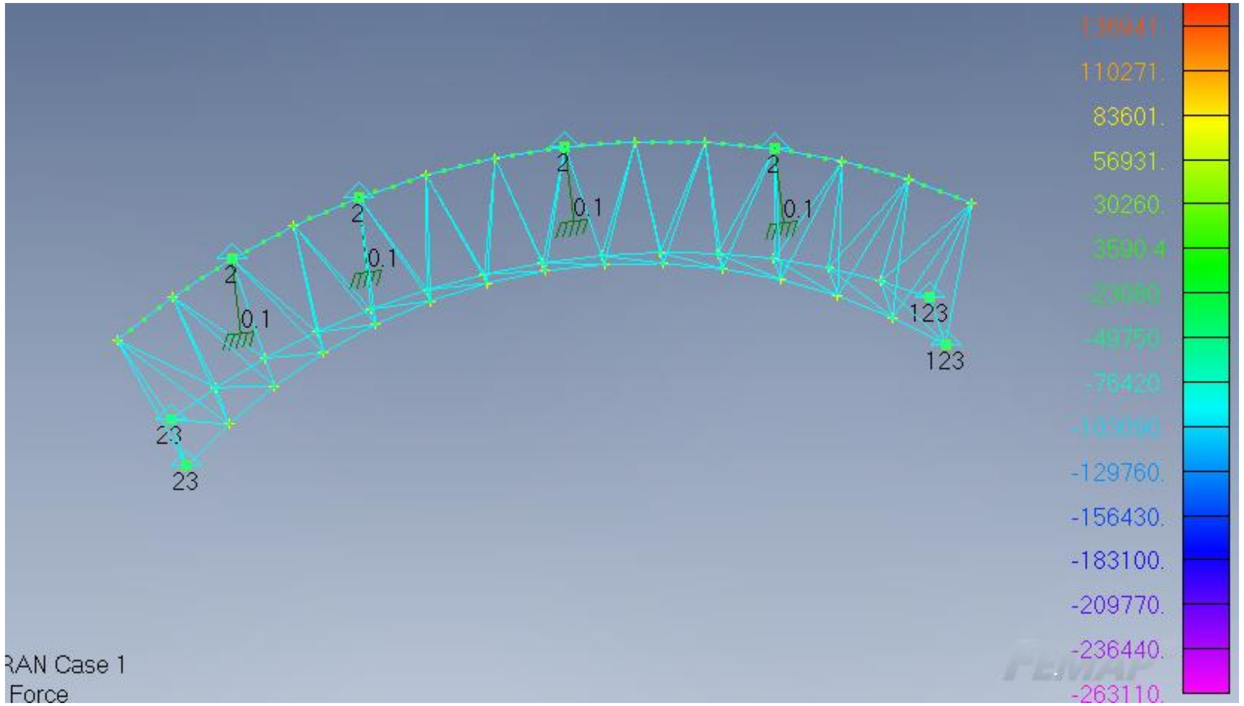
Lab 6 Planar (2D) Trusses from Geometry



Modeled a planar aluminum truss from geometry and assigned different rod properties based on cross-sectional area.
Post-processed the results to compare the deformed shape

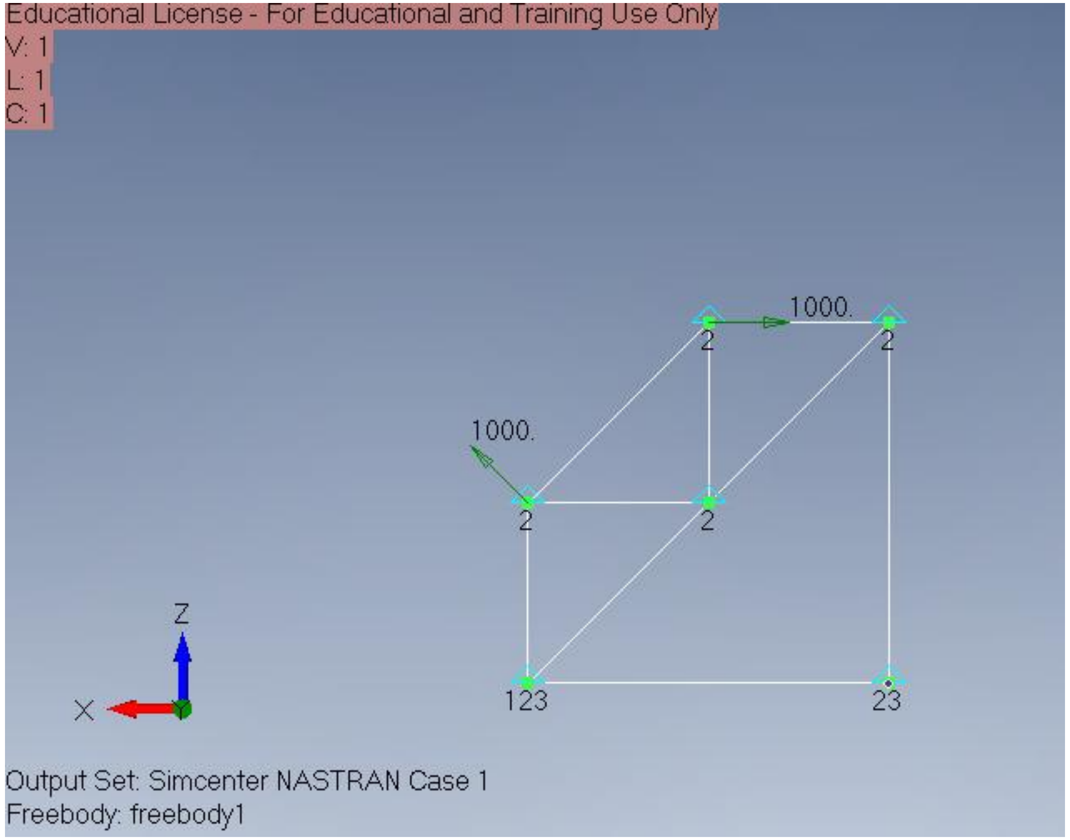
Lab 7 3D Truss — Geometry Imported from CAD

Imported a SolidWorks point-based 3D truss geometry into FEMAP and built the truss model using rod elements.

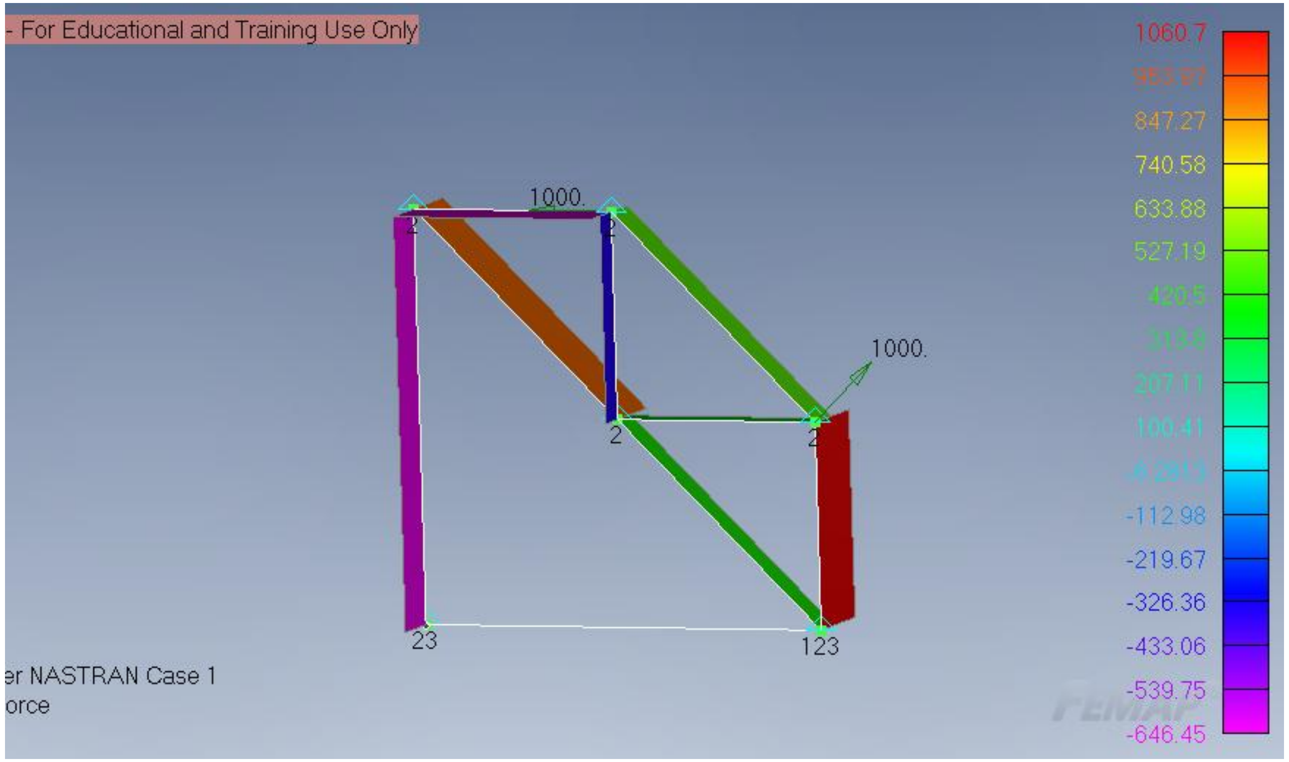


Attempted to run the FEA to compare deformed and undeformed shapes, support reactions, axial member forces, and stress locations.

Lab 9 Nastran Input Deck of a 1D Rod

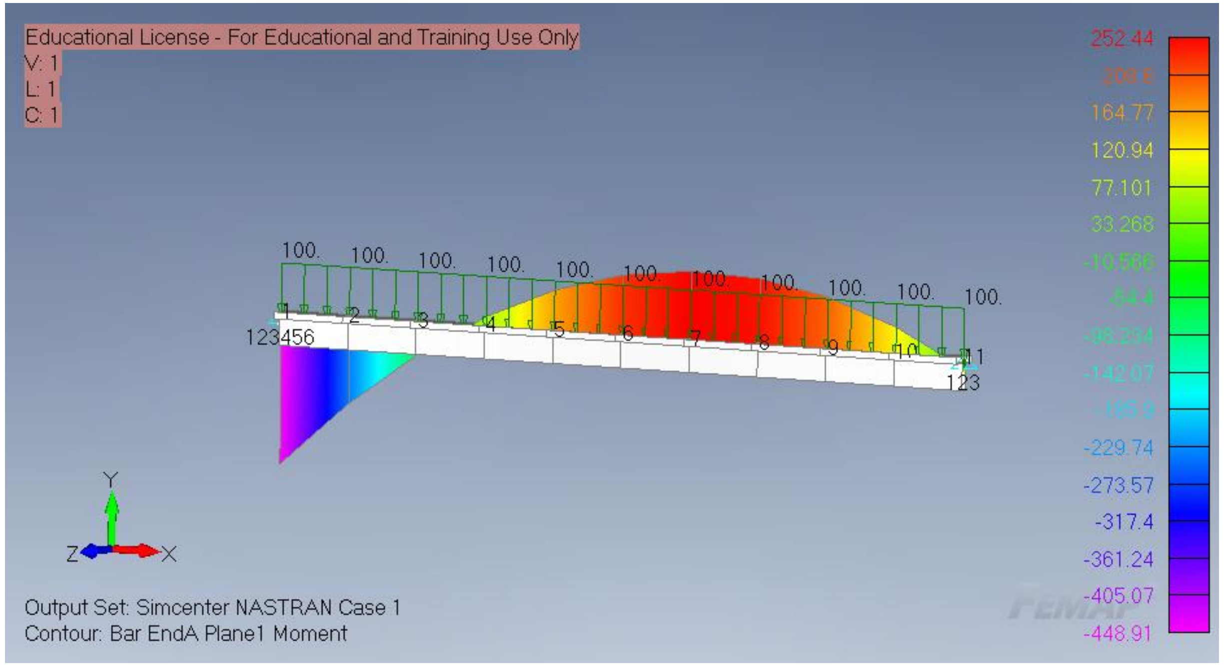


Created a planar truss model by modifying a Nastran rod input deck instead of building the geometry directly in FEMAP.

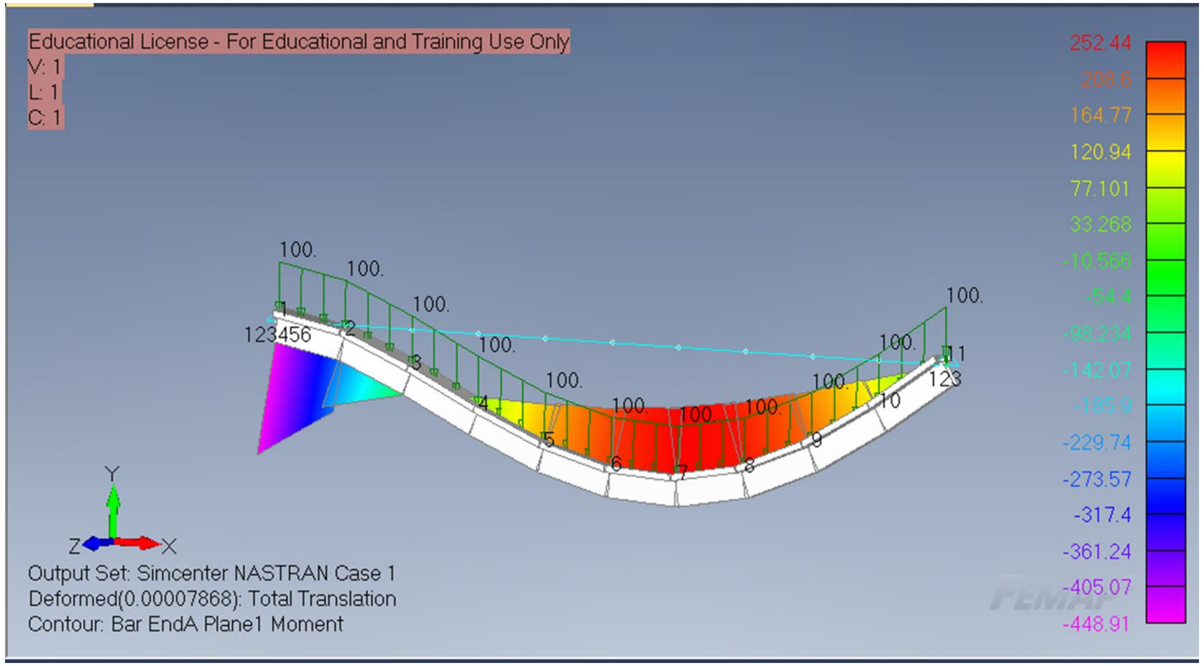


Post-processed the FEMAP results to evaluate displacement, internal member loading, stress, and reaction forces.

Lab 10 FEA of a 1D Bar (Beam)

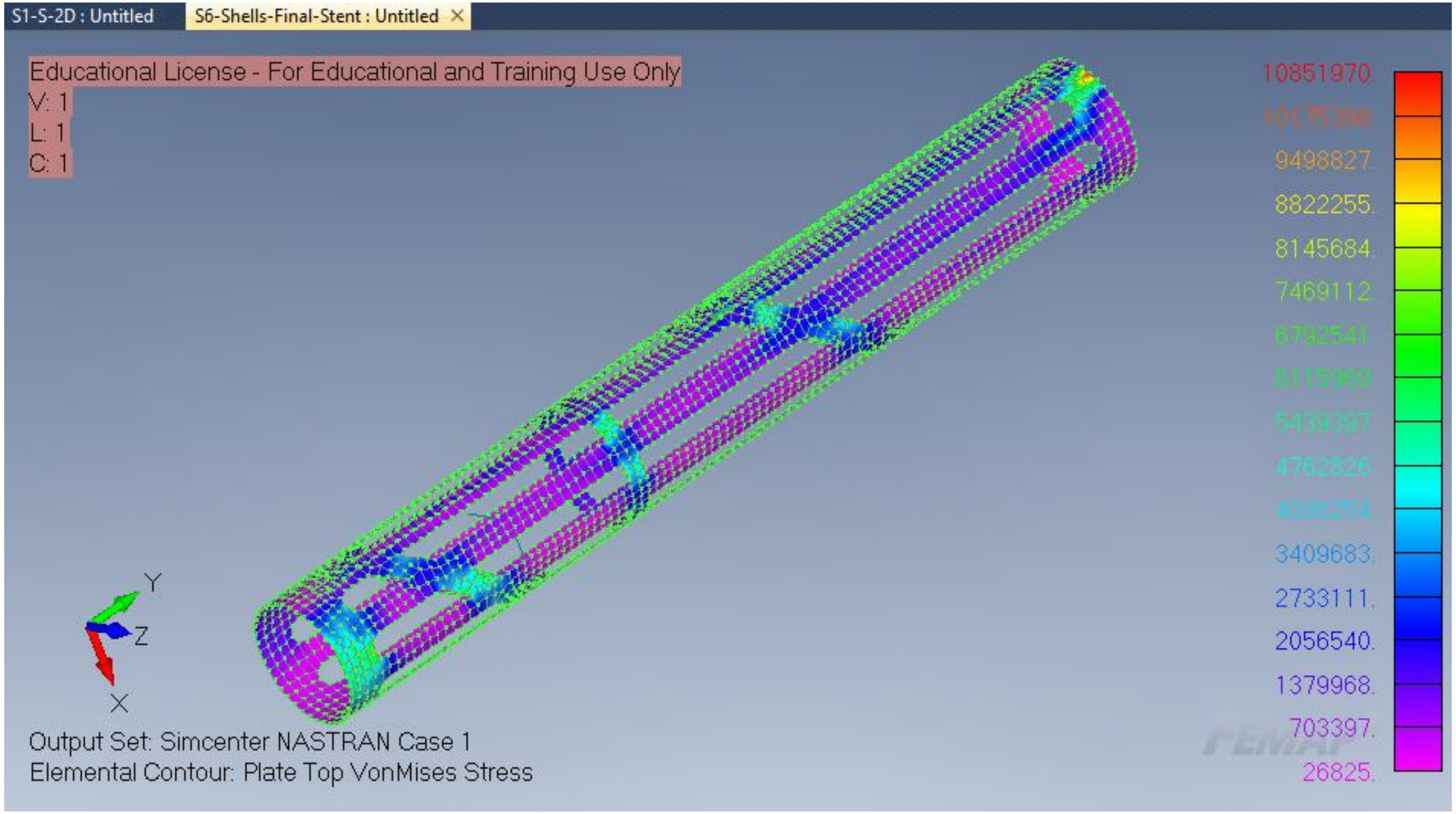


Created a 1D beam FEA model of an aluminum T-beam with assigned material properties, beam cross-section, supports, and distributed loading.



Ran a static analysis and reviewed the deformed shape, bending moment, and stress results to evaluate the beam response.

Lab 11 Nastran Input Deck for a Stent (Shell)

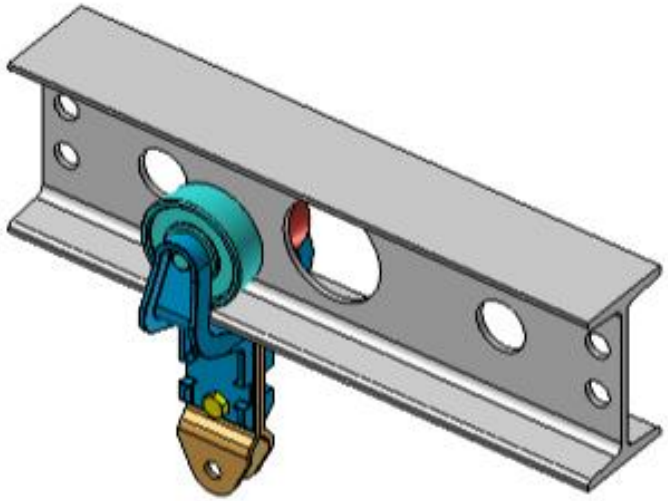


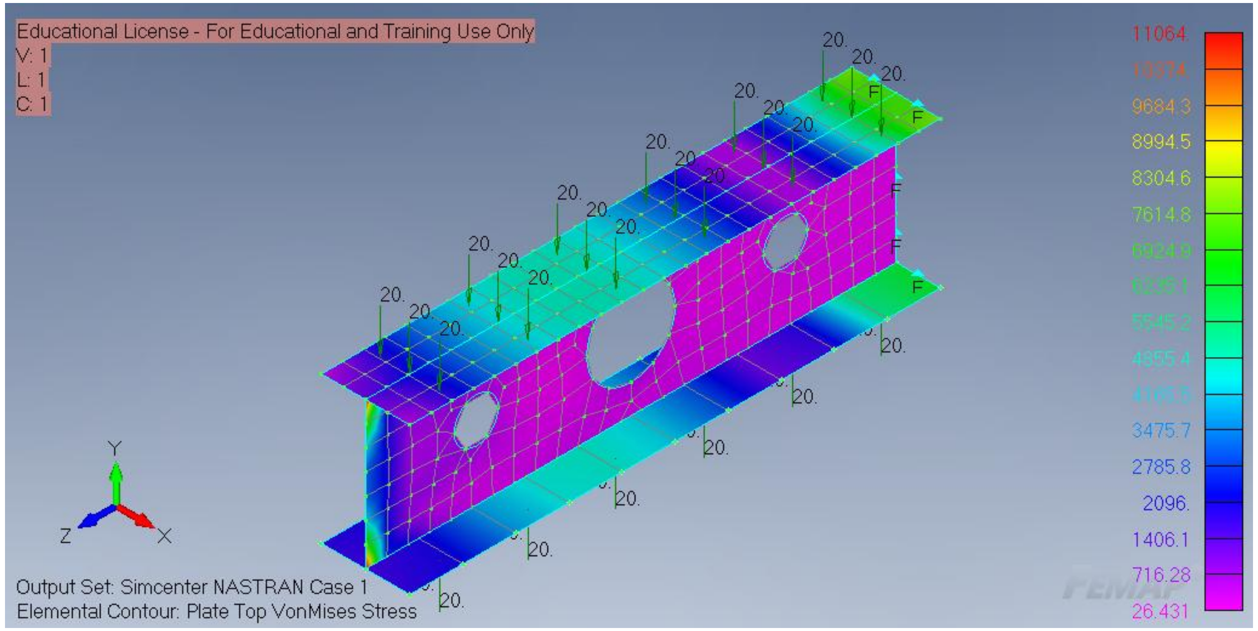
Transformed the flat 2D stent pattern into a cylindrical shell mesh using a Nastran input deck

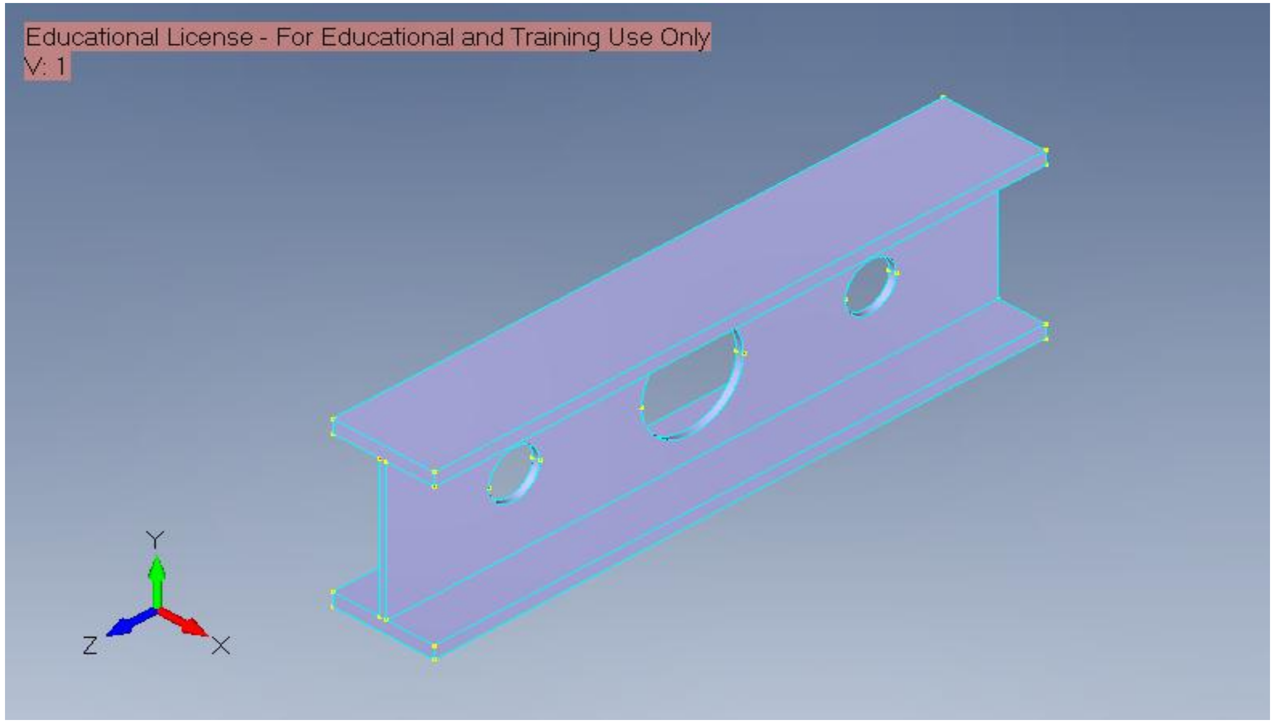
Lab 12 3D CAD Model — Defeaturing and Mid-Surfacing

Imported a SolidWorks structural rib model into FEMAP and simplified the geometry by removing small holes, chamfers, and fillets.

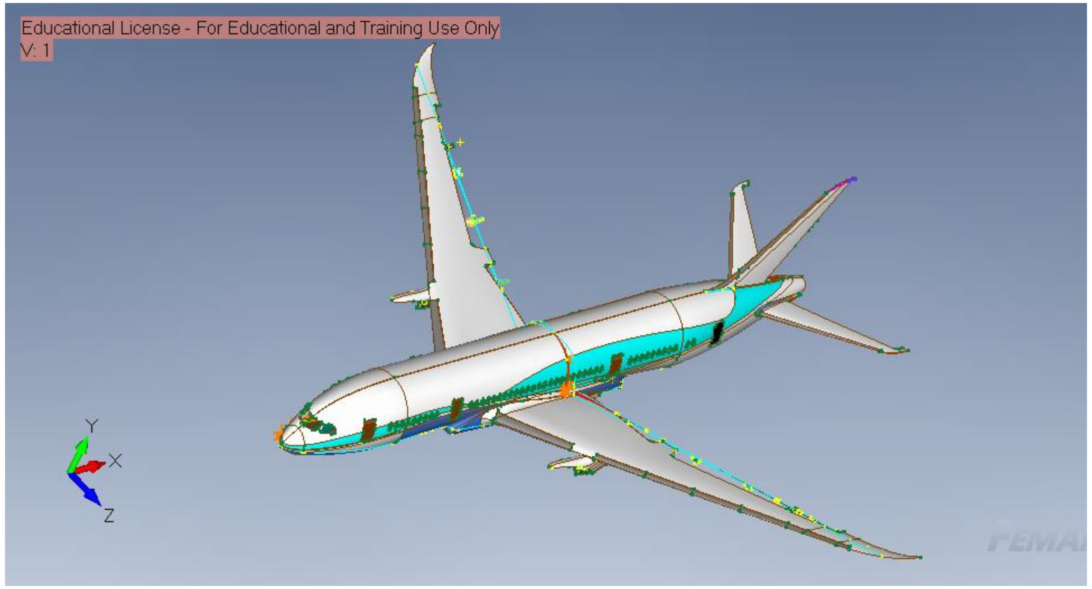
Used the shell model to apply loads, constraints, and evaluate results.

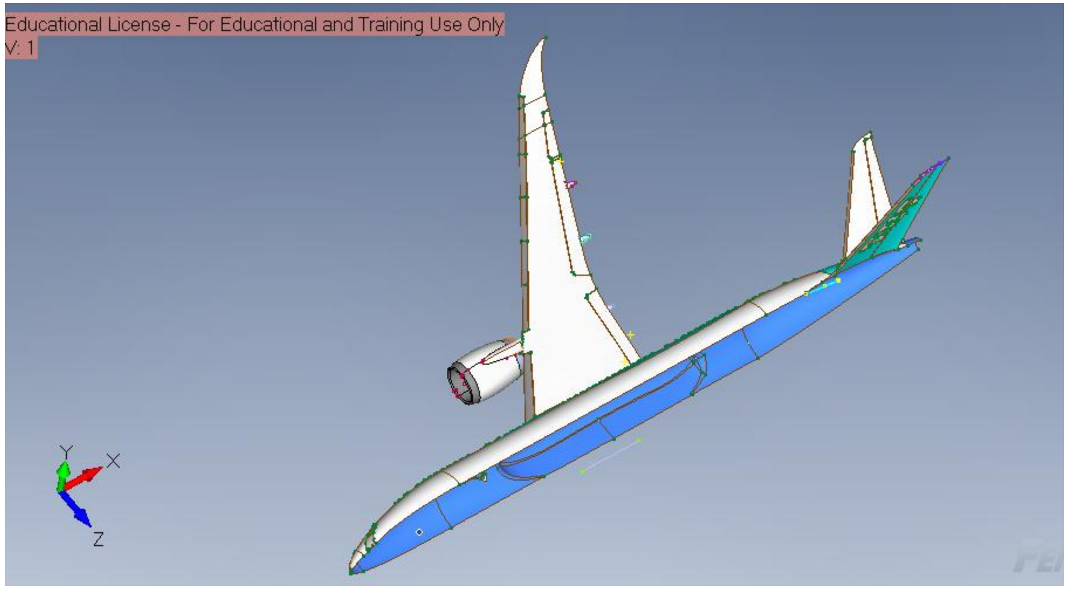




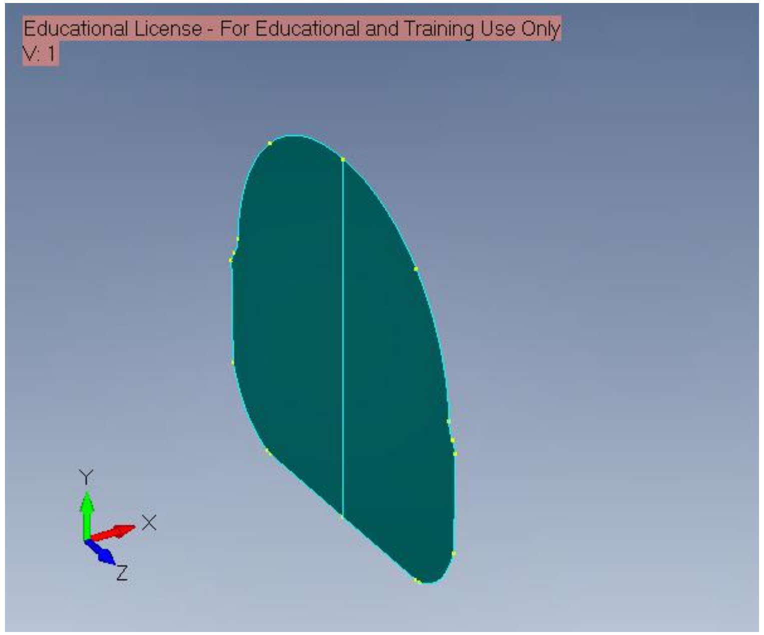


Lab 14-15 3D CAD Model — Geometry Defeaturing





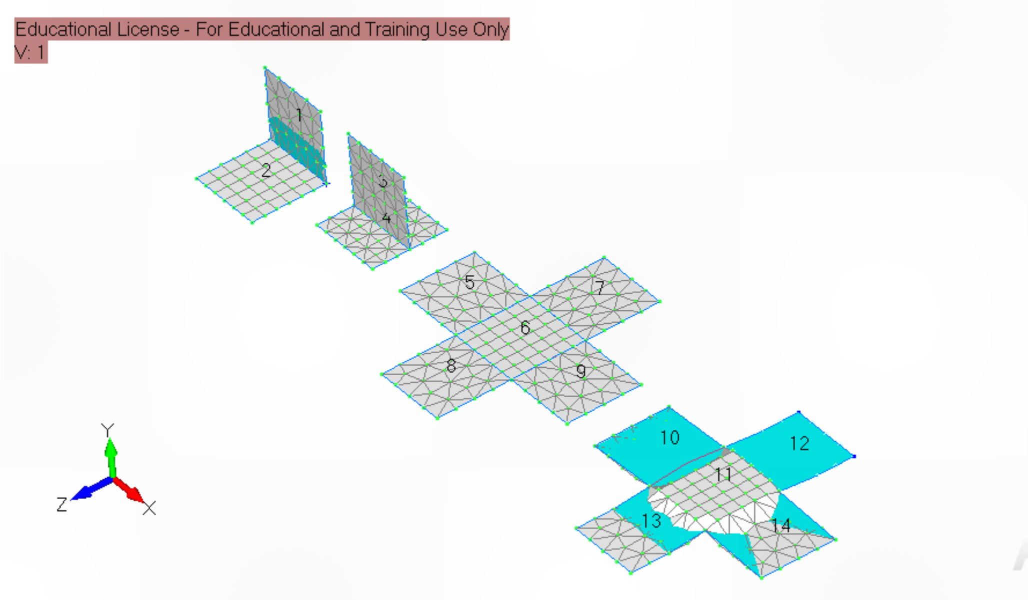
Performed geometry defeaturing on the Boeing 787 model by removing engines, flaps, logos, and other non-essential details.
Cleaned and stitched the mirrored remaining surfaces to reduce gaps

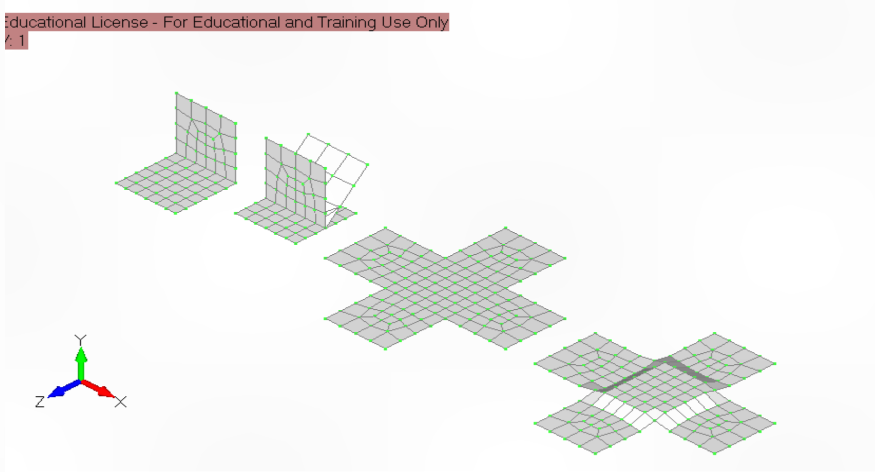


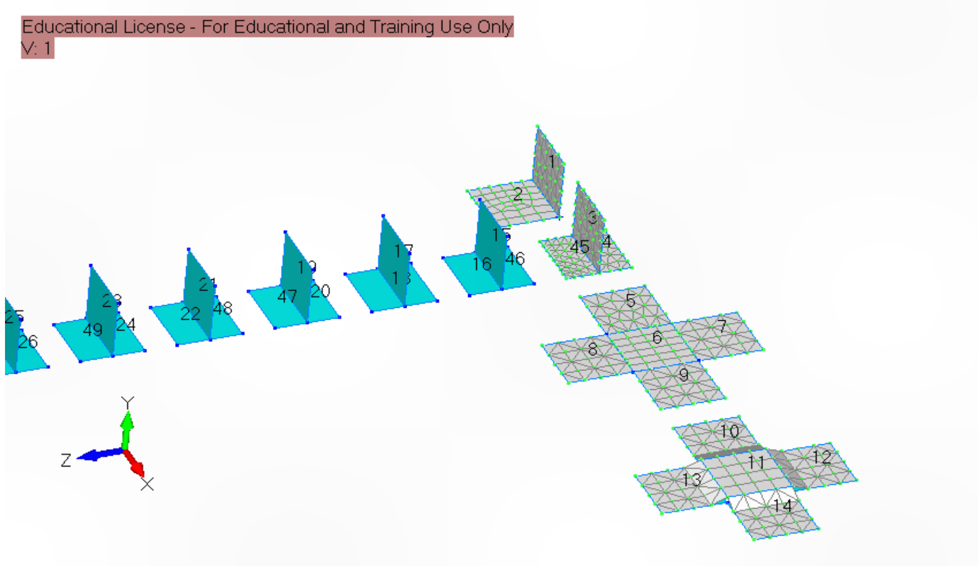
Lab 16 Shell Meshing — Preventing Disjoint Meshes

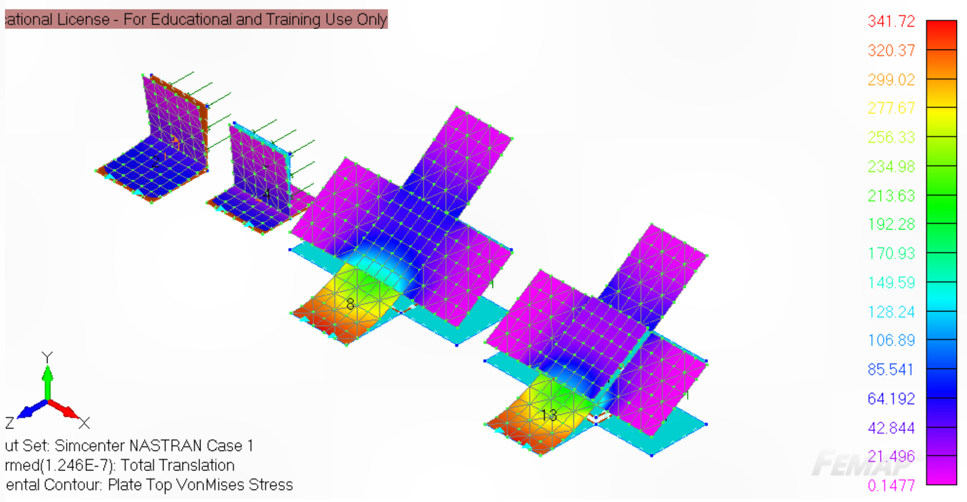
Practiced two mesh-connectivity methods: stitching adjacent surfaces and merging duplicate nodes along shared free edges.

The goal was to create continuous shell meshes without unwanted gaps between surfaces.





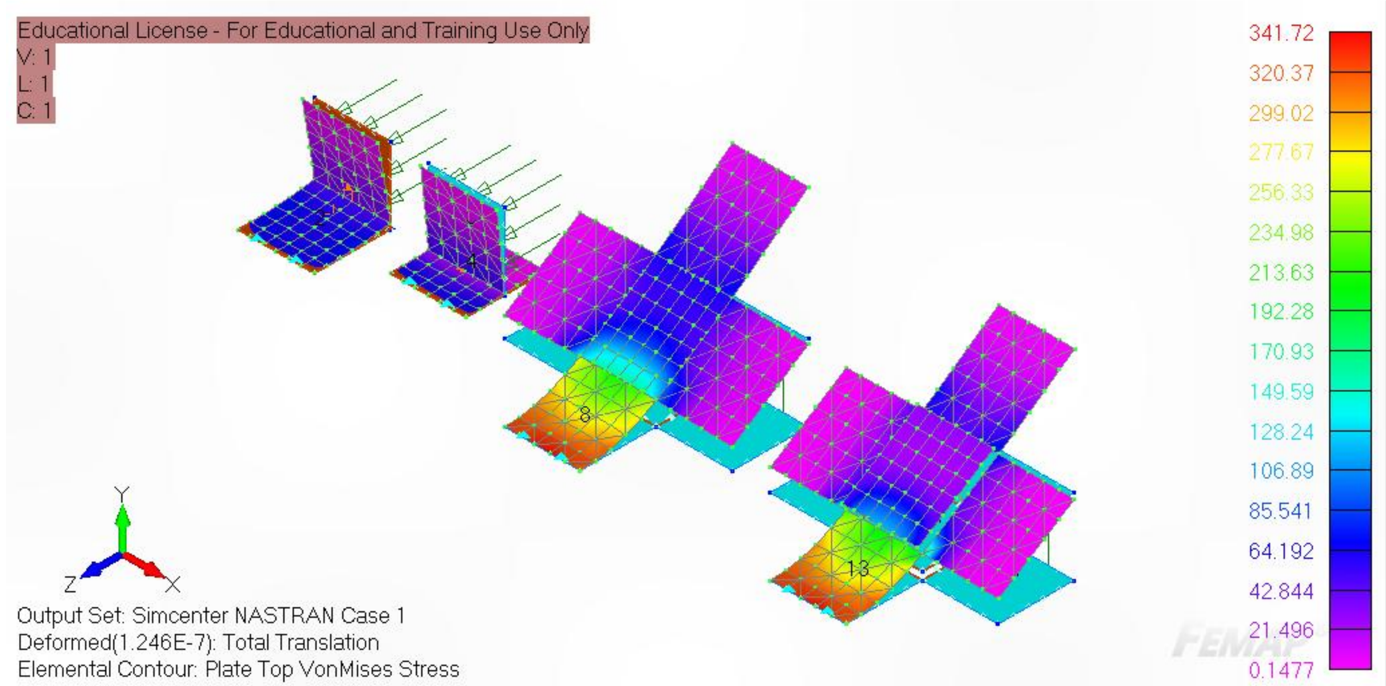


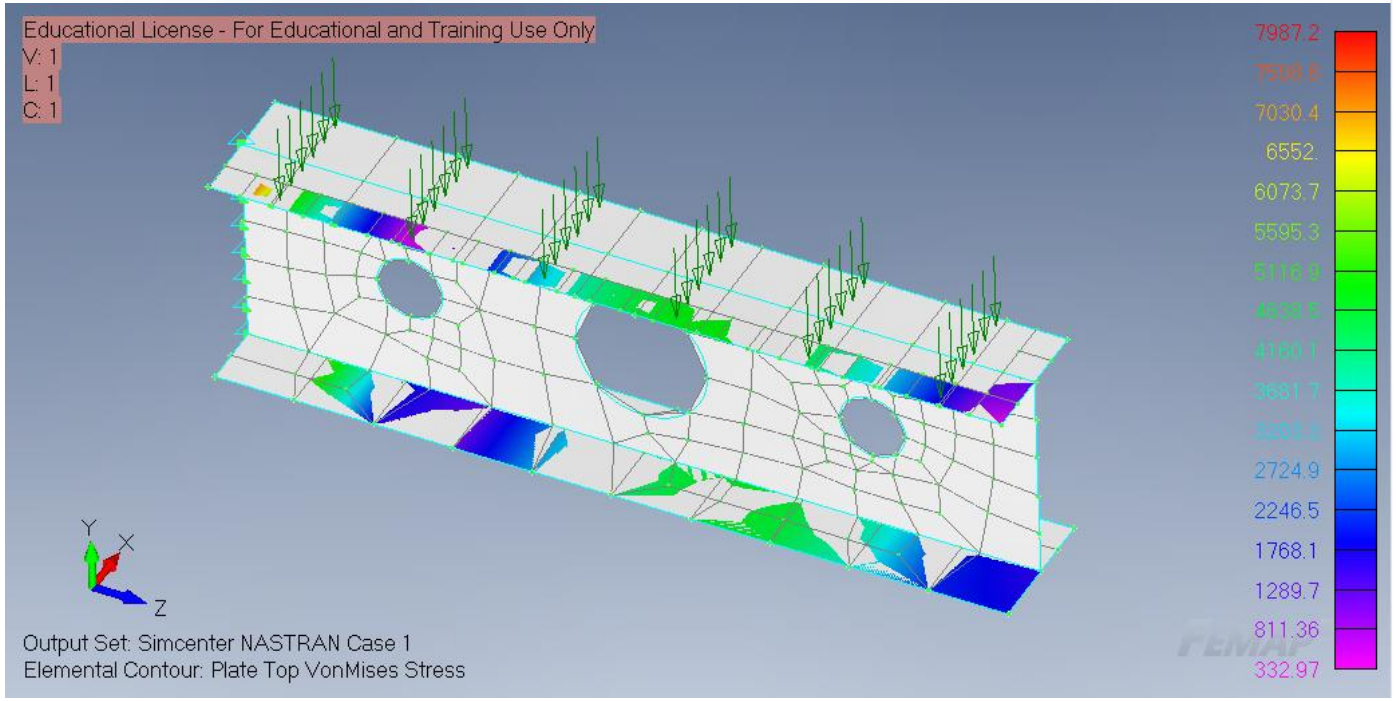


Lab 17 Shell Meshing — Preventing Disjoint Meshes

Practiced shell meshing techniques to prevent disjoint meshes by removing free edges and merging duplicate nodes.

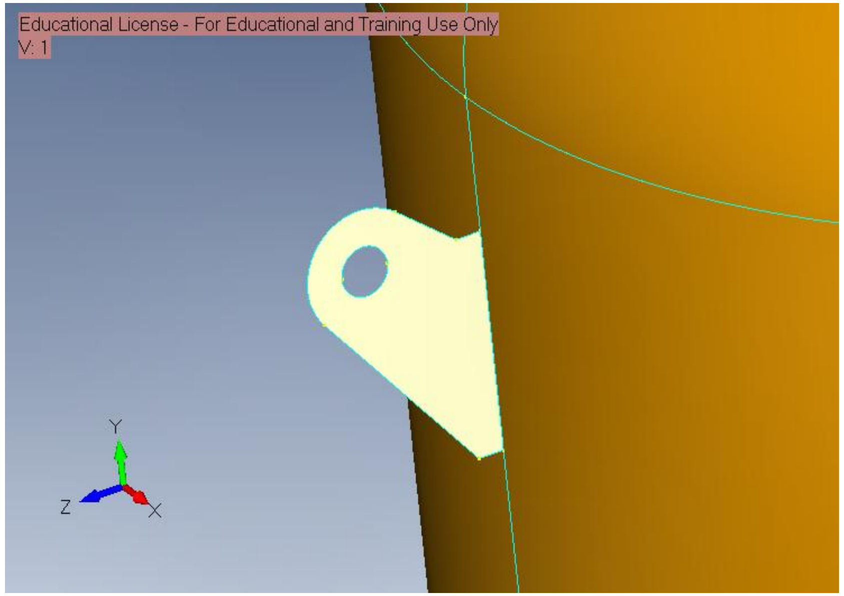
Defeatured and mid-surfaced a structural rib model to create a thin-walled shell.

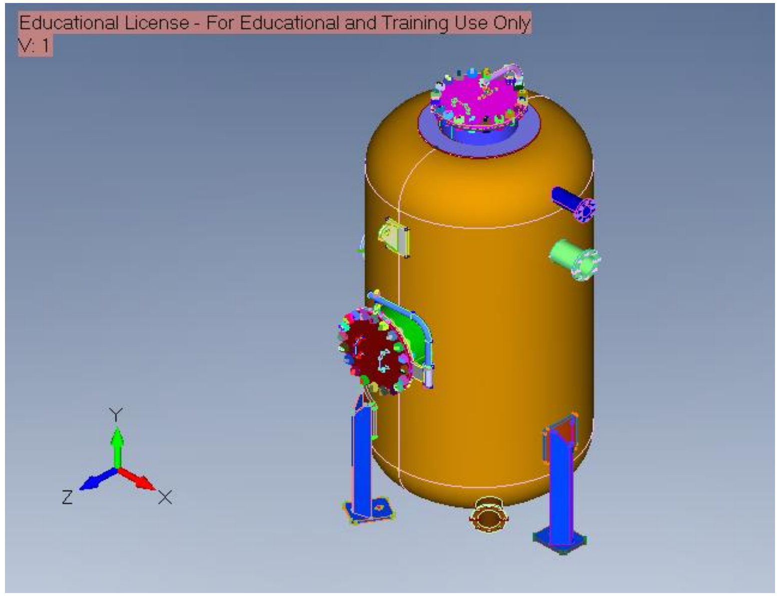


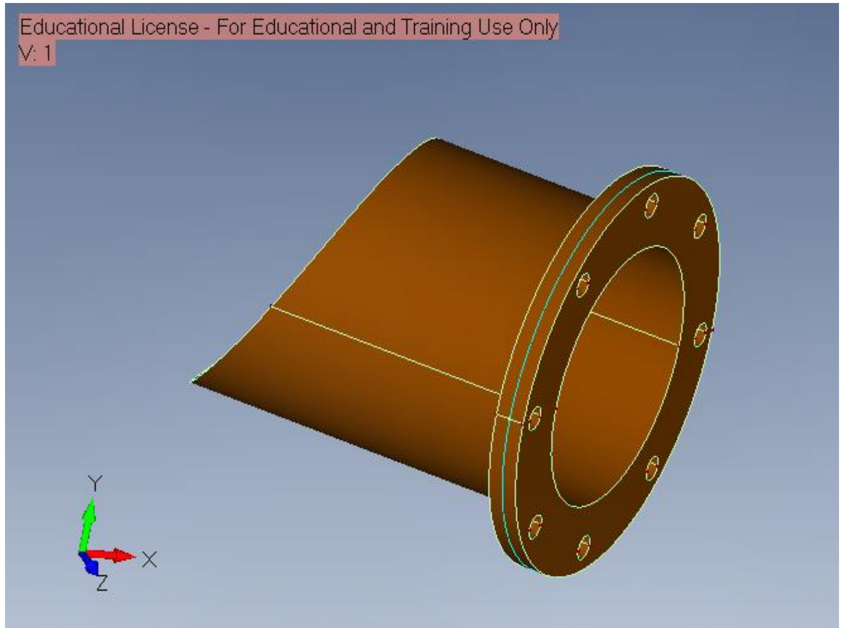


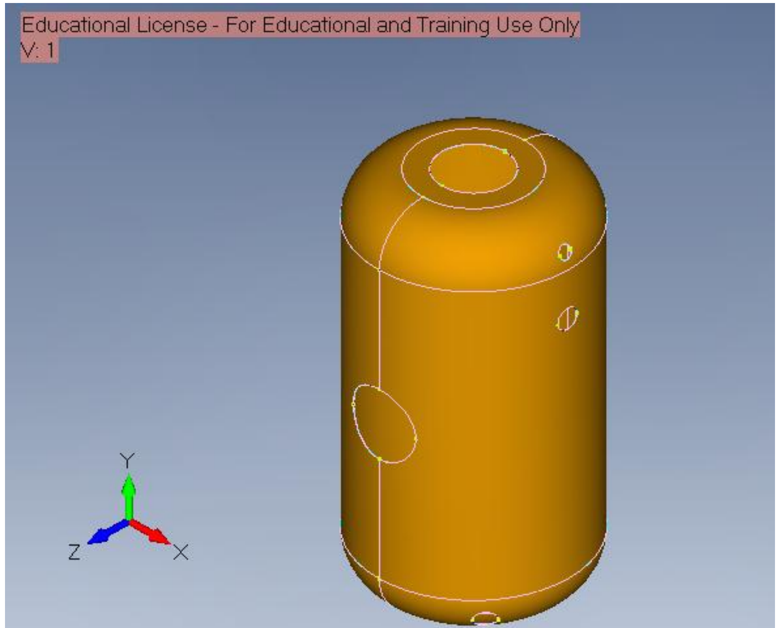
Lab 18 FEA of a Pressurized Cylindrical Tank

Defeatured and mid-surfaced the cylindrical tank geometry to create a shell-element model of the pressure vessel.





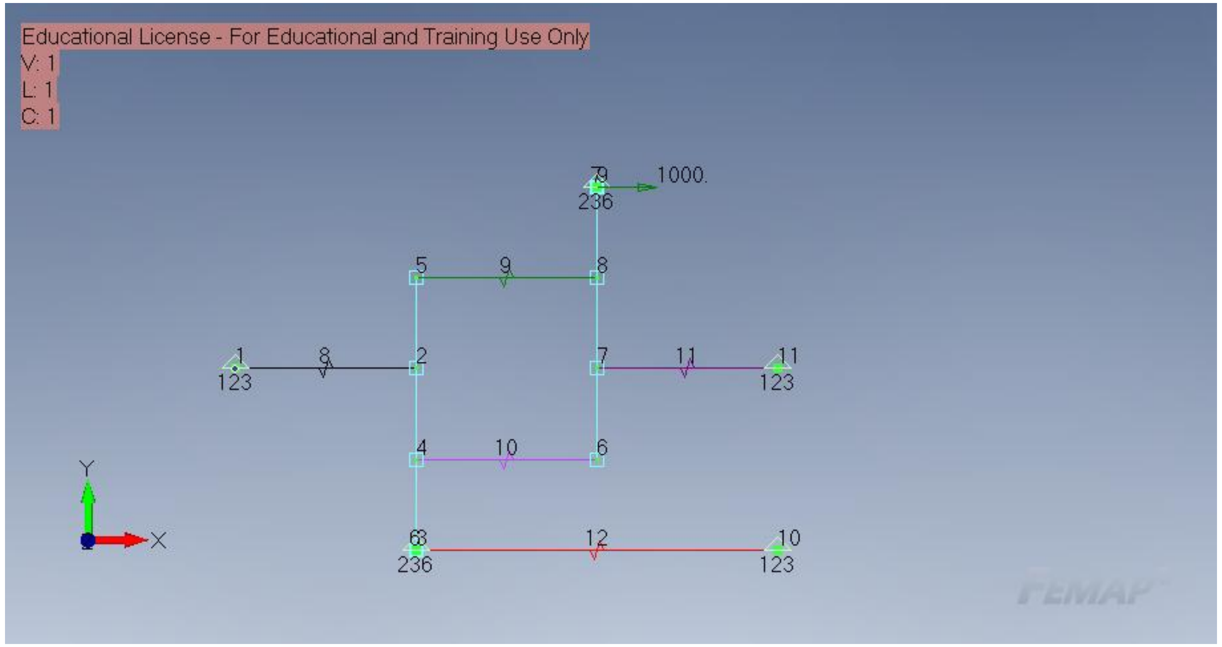


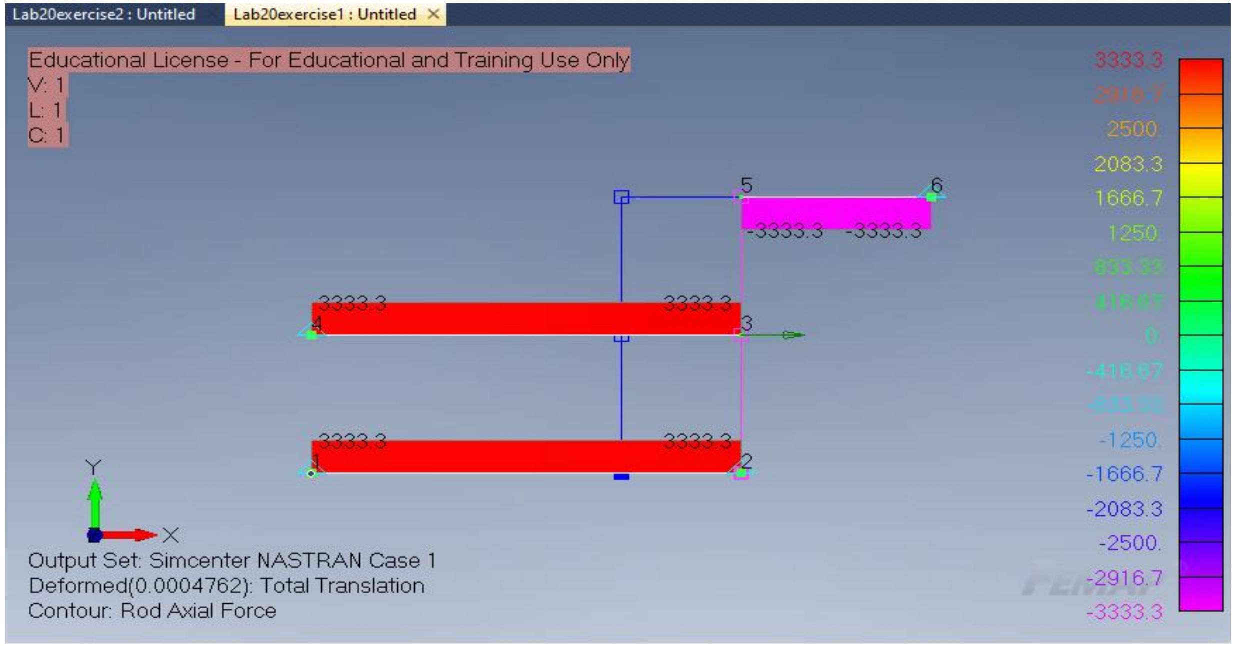


Lab 20: The Rigid-Body Element
RBE2

Modeled rigid-body connections between rods and springs using RBE2 elements to represent rigid bars, applied force at nodes.

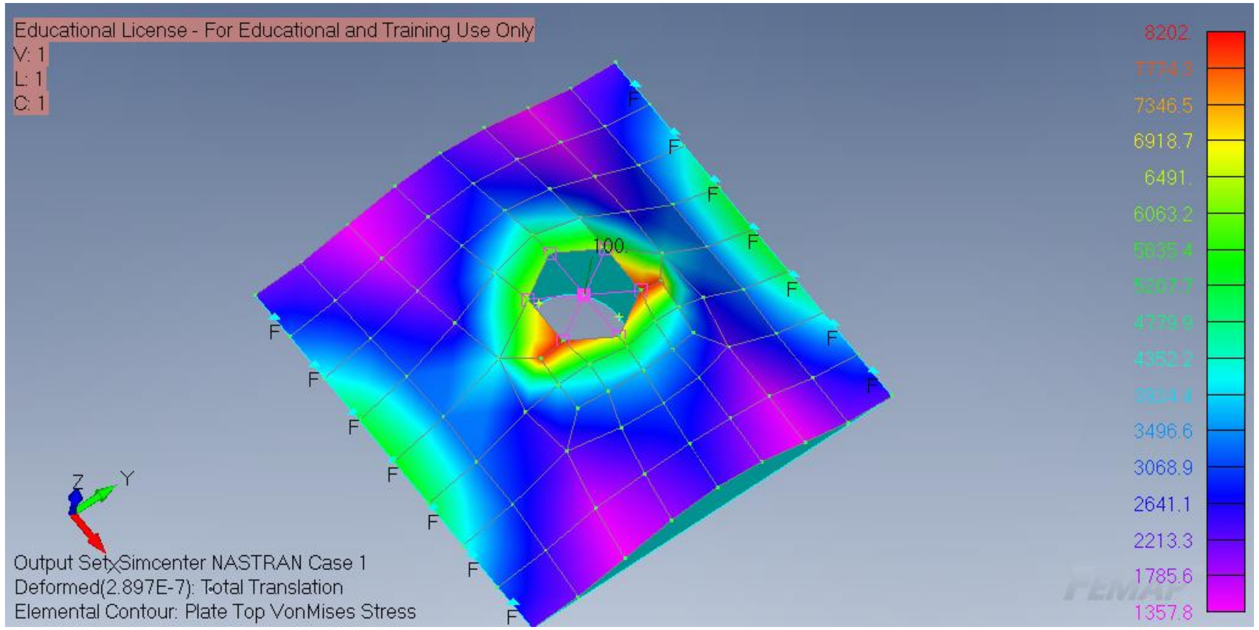
Ran the analysis to determine nodal displacements.



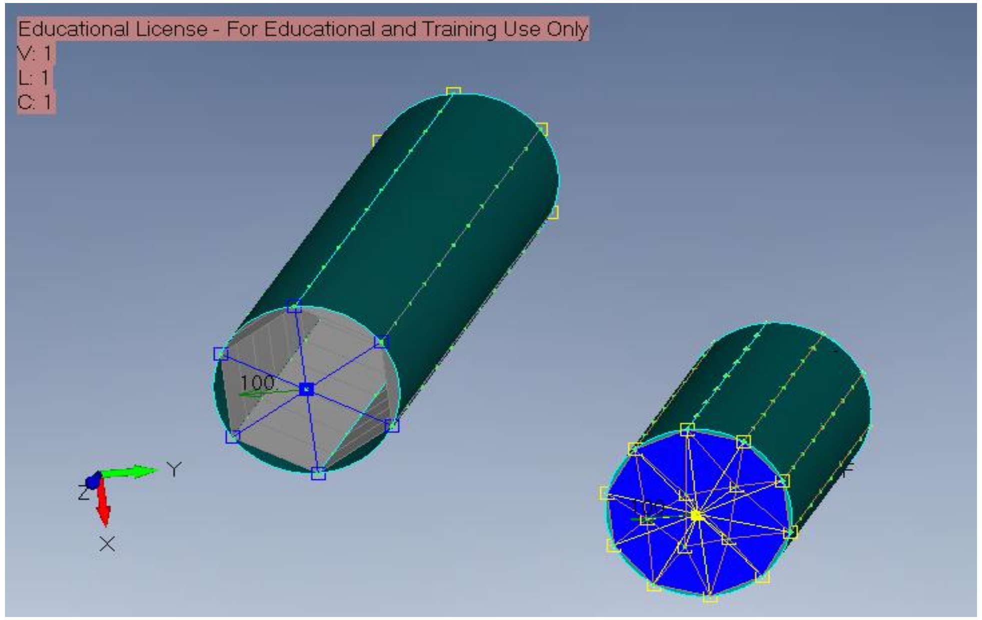


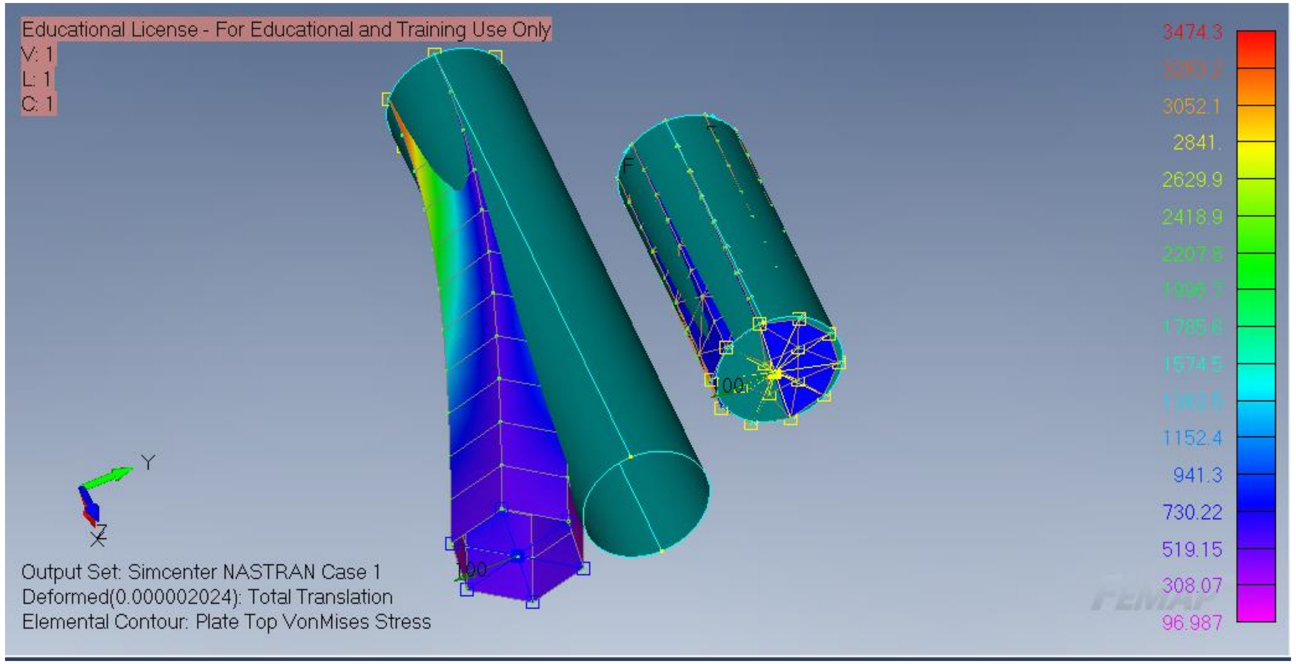
Lab 21 Stress Analysis of Remotely Loaded Members

Created remotely loaded plate and cylinder models using RBE2 rigid-body elements to transfer loads from a master node to boundary nodes.



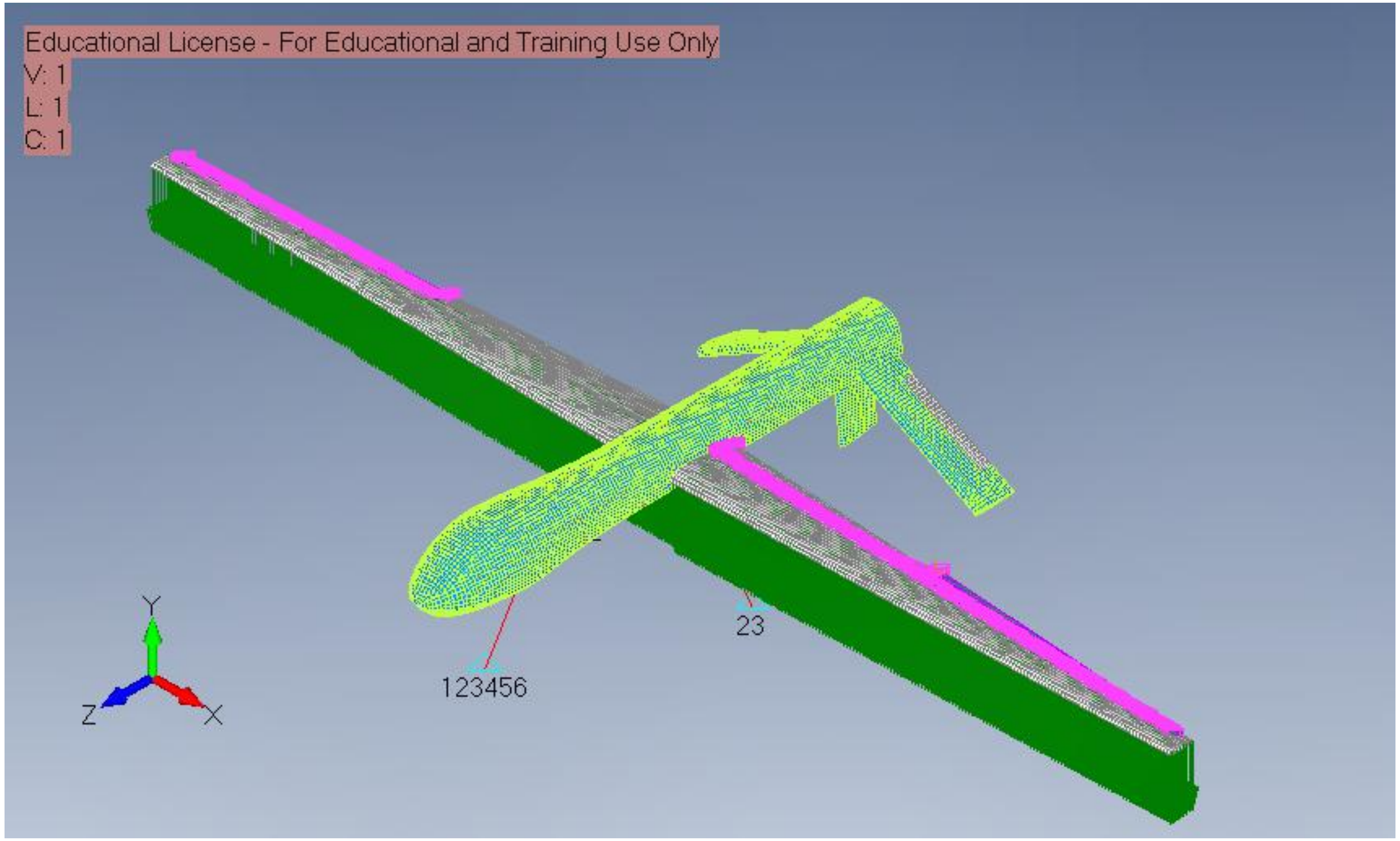
Applied remote loads to hollow and solid cylinder models using rigid-body elements at the free end.





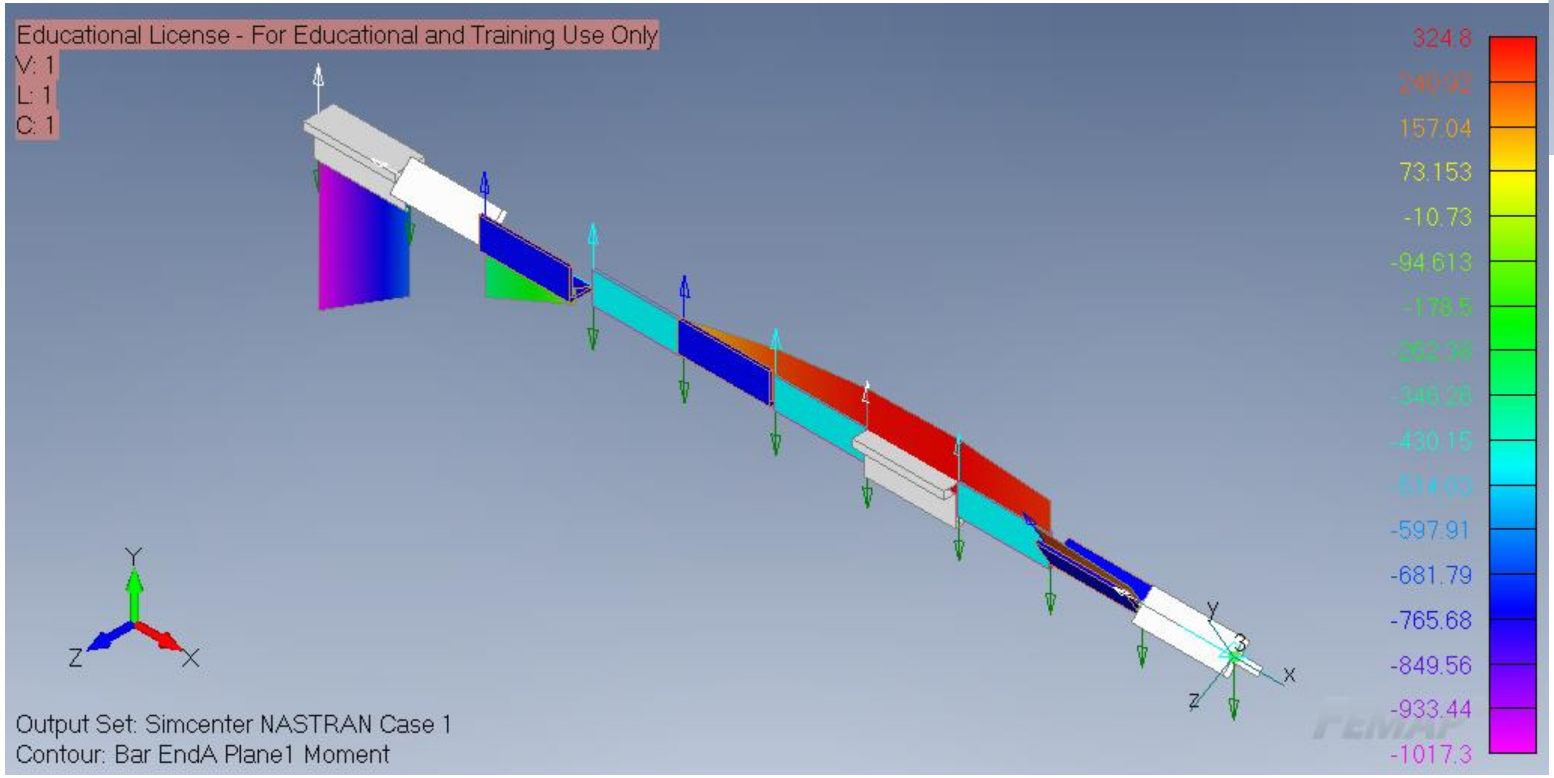
Lab 22 Static FEA of a U.A.V.

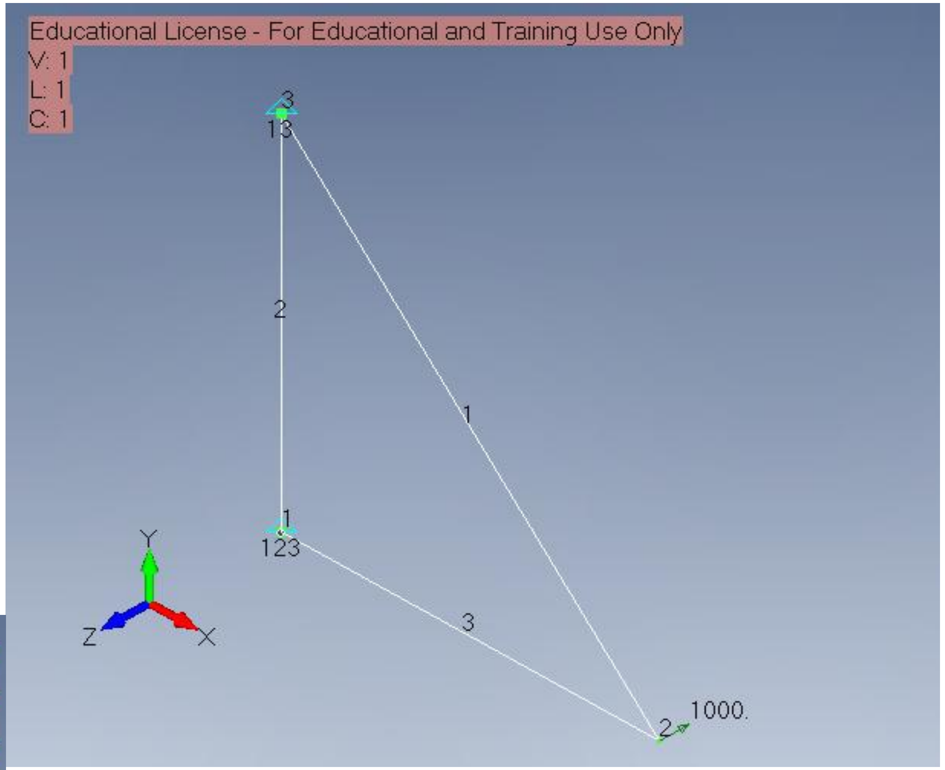
Prepared the UAV FEMAP model with mesh regions, wing constraints, and applied loading locations.



Lab 23 FEA of Planar Frames

Modified the previous planar truss model by replacing rod elements with bar elements to create rigid frame joints.

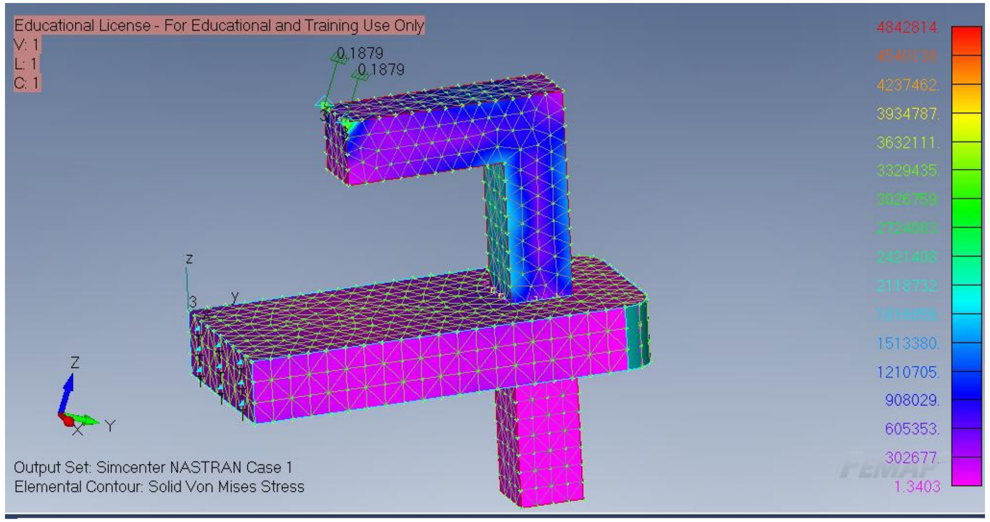




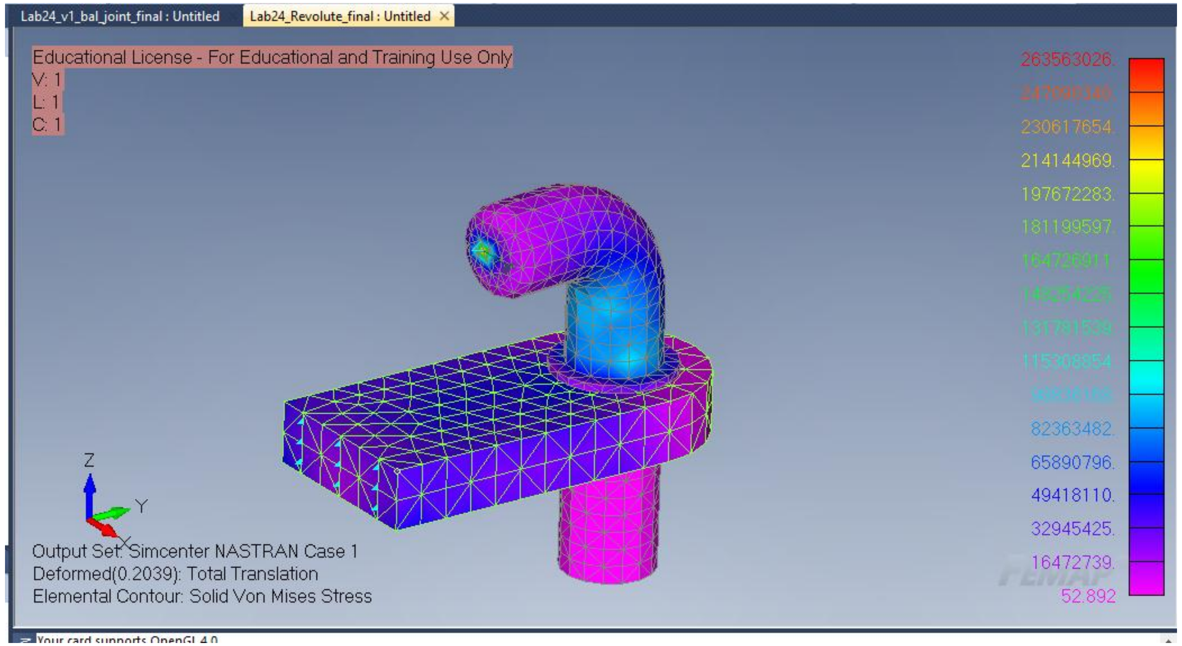
Modified the previous planar truss model by replacing rod elements with bar elements to create rigid frame joints.

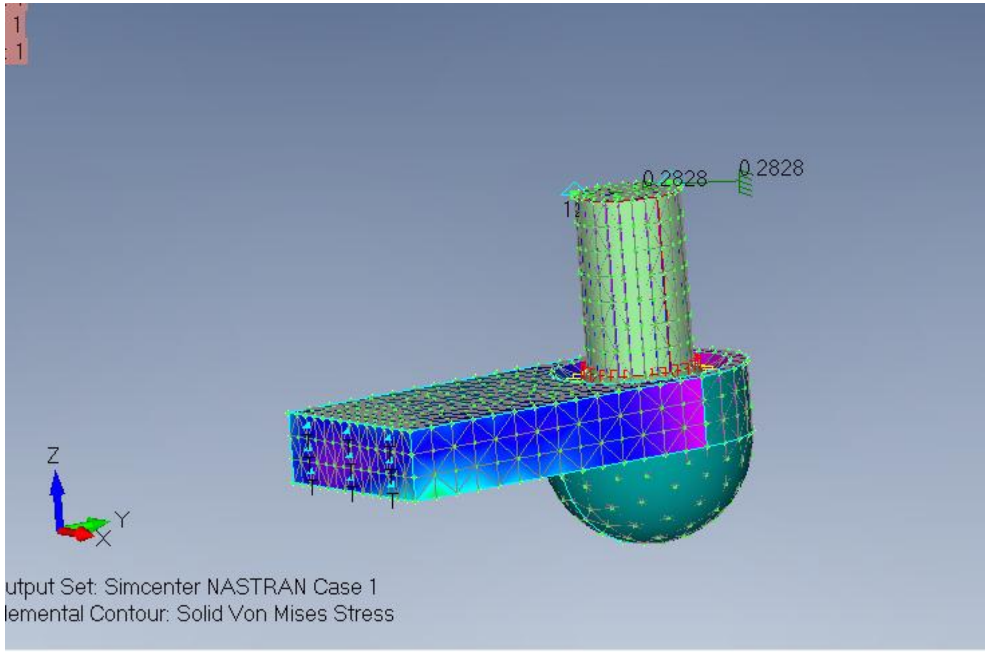
Lab 24: FEA using RBE2's

I used RBE2 elements to connect the pin and slot regions, then added a spring element to represent the sliding joint motion.



RBE2 elements to connect the pin and hole regions to reference points, then added a spring element to represent the joint behavior.





Created RBE2 connections for the ball and socket regions and connected them using a spring element. This setup represents a ball-and-socket type joint, where rotational motion can occur about multiple axes while the connected parts remain joined at the joint center.